Fourier Decay of Rvachev's Up-Function

Exact products, pressure spectra, Cesàro gauges, fluctuations, peaks, and valleys

A human-readable synthesis for the `ProveIt` repository

31 August 2026

Abstract

Let

$$\Phi(x) = \widehat{\mathrm{up}}(x) = \prod_{h=0}^{\infty} \operatorname{sinc}\Big(\frac{\pi x}{2^h}\Big), \qquad \operatorname{sinc} z = \frac{\sin z}{z},$$

be the Fourier transform of Rvachev's compactly supported up-function. This article consolidates the original 2018 question, eight successive research reports, a gentle guide, two adversarial audits, their verification scripts, and the relevant lacunary-products literature into one self-contained synthesis. The main conclusion is that the phrase "the decay rate" has no unqualified meaning. There is a universal deterministic factor

$$\exp\Big(-\frac{(\log x)^2}{2\log 2}\Big),$$

but the next power of x depends on whether amplitude means a typical value, an arithmetic mean, an RMS value, a pointwise upper envelope, or the largest value in a dyadic annulus.

An exact dyadic factorization reduces the remaining numerator to a lacunary sine cocycle for the doubling map. Its L^q pressure yields a complete shell spectrum. In particular, with $E_\kappa(x) = x^{-\kappa}\exp(-(\log x)^2/(2\log 2))$, the four principal powers are

$$\kappa_0 = 3.1514961294723187980\ldots, \qquad \kappa_1 = 2.7480700148713326739\ldots,$$
$$\kappa_2 = 2.6514961294723187980\ldots, \qquad \kappa_\infty = 2.3590148791117407073\ldots.$$

The first, third, and fourth decimals come from exact closed forms; the displayed κ_1 digits are high-precision spectral computation, with a separate rigorous enclosure recorded below. They correspond respectively to geometric mean / almost-everywhere dyadic rays, arithmetic mean, root mean square, and the sharp pointwise upper envelope. The old conjectural power $\log_2(2\pi)$ is therefore exactly right for RMS decay and wrong for the other readings. No universal power of $\log x$ repairs the mismatch.

The metric fluctuation is also settled. For $S_n(t) = \sum_{j<n} \log|2\sin(\pi 2^j t)|$, an exact covariance computation gives

$$\int_0^1 S_n(t)^2\,dt = \frac{\pi^2}{4}n - \frac{\pi^2}{3}(1-2^{-n}).$$

Equivalently, S_n is a coboundary perturbation of the reverse martingale-difference sequence generated by $\log|\tan \pi t|$. Hence the correct law-of-the-iterated-logarithm constant is $\pi/2$ with denominator $\sqrt{2n\log\log n}$, or $\pi/\sqrt{2}$ with denominator $\sqrt{n\log\log n}$. The factor π printed in the 2017 lacunary-products theorem is too large by $\sqrt{2}$; the source of the discrepancy is the missing $1/2$ in the L^2 norm of the cosine basis.

For the literal Cesaro question, the arithmetic-mean gauge $g(x) = A_1 E_{\kappa_1}(x)$, where $A_1 = 0.091266124131588\ldots$, gives shell means tending to one and solves the requested two-sided boundedness condition. It also gives convergence at dyadic endpoints. At unrestricted endpoints a nonconstant dyadic-mantissa profile remains, governed by a continuous singular eigenmeasure of an explicit transfer operator. We prove that no gauge with a stabilizing within-shell shape (including powers and iterated logarithms, regularly varying factors, and fixed log-periodic

corrections) can remove this profile. Unrestricted scale-dependent gauges are not ruled out by that theorem; in fact, a shell-adaptive positive real-analytic decreasing gauge is constructed below and gives the literal unrestricted Cesaro limit. The multiplicatively natural logarithmic Cesaro mean does converge for all endpoints, with normalizing constant $A_1^{\log} = 0.09386063575678126354\ldots$.

Finally, the raw annular maximum $\mathfrak{M}_k = \max_{2^k \le x \le 2^{k+1}} |\Phi(x)|$ has a sharper boundary-layer law:

$$\log \mathfrak{M}_k = -\frac{\log 2}{2}k^2 - (\log 2)\kappa_\infty k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\log 2} + O((\log\log k)^2),$$

where $c = (\log 2)\pi^{-1}\sqrt{3/2}$ and W is the principal Lambert function. Thus the largest point in a shell approaches its left boundary and carries a second log-normal factor on the $\log\log x$ scale. This is normalized at the shell's left endpoint and is fully compatible with the exact fixed-mantissa ray that is sharp relative to E_{κ_∞} at its own argument. In contrast, the unique lobe maxima immediately after powers of two form explicit deep valleys with leading coefficient $-1/\log 2$, twice the generic log-square coefficient. We also correct and prove the original reciprocal-Gamma and q-Pochhammer tail formula. Exact proofs, imported classical analytic inputs, rigorous certificates, and numerical evidence are distinguished throughout.

Contents

1 The question and the shape of the answer

Rvachev's up-function is the even C^∞ probability density supported on $[-1,1]$ that is closely related to the Fabius distribution function. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-2\pi \mathrm{i} t\xi}\,\mathrm{d}t,$$

its transform is the entire function

$$\Phi(z) = \widehat{\mathrm{up}}(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right). \tag{1.1}$$

This formula is classical; see, for example, Arias de Reyna's systematic treatment [1] and the Fabius/Rvachev development in [13].

The motivating questions in [11, 12] ask for the amplitude of the oscillations and, more specifically, for a positive elementary function g such that an average of $|\Phi|/g$ tends to one or at least stays between two positive constants. A working draft in the `ProveIt` repository [14] proposes a log-normal factor together with the power $x^{-\log(2\pi)/\log 2}$ and a possible power of $\log x$. The log-normal factor is indeed fundamental, but the rest of the conclusion depends on which norm or averaging operation is intended.

Here is a cleaned statement of the two averaging questions. Fix $t_0 > 0$ and ask for a positive, eventually decreasing, real-analytic gauge g on $(0,\infty)$ such that either

$$\frac{1}{t-t_0}\int_{t_0}^{t} \frac{|\Phi(x)|}{g(x)}\,\mathrm{d}x \longrightarrow 1 \qquad \text{as } t\to\infty, \tag{Q2}$$

$$0 < \liminf_{t\to\infty} \frac{1}{t-t_0}\int_{t_0}^{t} \frac{|\Phi(x)|}{g(x)}\,\mathrm{d}x \le \limsup_{t\to\infty} \frac{1}{t-t_0}\int_{t_0}^{t} \frac{|\Phi(x)|}{g(x)}\,\mathrm{d}x < \infty. \tag{Q3}$$

The original wording also asked about monotonicity of higher derivatives. That optional request is ambiguous unless one specifies alternating signs and whether the tail may depend on the derivative order. We use the precise order-by-order formulation in Theorem 9.4 and state the limitation of the adaptive gauge after Theorem 10.3.

The final map is unambiguous. The natural gauge $g_1 = A_1 E_{\kappa_1}$ solves Q3 and Q2 along dyadic endpoints, but has a nonconstant all-endpoint phase profile. No stabilizing or bounded-distortion correction solves Q2. A shell-adaptive analytic decreasing gauge does solve Q2, while the fixed natural scale also gives an unrestricted limit after replacing additive Cesaro averaging by logarithmic averaging.

There are three separate obstructions to a single pointwise answer.

1. The transform has a zero at every nonzero integer, so no asymptotic $|\Phi(x)| \sim g(x) > 0$ can hold on the full real ray.
2. After the deterministic denominator is removed, what remains is a Birkhoff product for the doubling map. Its geometric mean, arithmetic mean, quadratic mean, and supremum have different exponential growth rates.
3. Even the unique maxima between consecutive zeros possess exceptional subsequences. The binary period-two ray realizes the sharp global-envelope rate exactly up to the constant C_*, whereas the extrema in the first few intervals after a power of two are exponentially smaller on the n^2 scale.

Thus the correct answer is a *spectrum of decay rates*, not one formula. A convenient common scale is

$$E_\kappa(x) := \exp\left(-\frac{(\log x)^2}{2a} - \kappa \log x\right) = x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2\log 2}\right), \qquad a := \log 2. \tag{1.2}$$

For fixed dyadic phase, the quadratic term in (1.2) comes from the product of the n denominators $2^1, \ldots, 2^n$. The linear term is where the lacunary numerator enters.

1.1 Main decay table

The following table summarizes the results proved below. In the finite-q rows, "normalized" means the normalized L^q norm on the dyadic shell $[2^{n-1}, 2^n]$.

Amplitude notion	numerator scale Λ	power κ in E_κ	conclusion
Geometric mean ($q = 0$) and typical dyadic ray	$1/2$	$\frac{3}{2} + \frac{\log \pi}{a}$	shell geometric mean is exactly constant after normalization; the same exponent holds for Lebesgue-a.e. fixed phase
Arithmetic mean ($q = 1$)	$\rho_1 = 0.6613226021\ldots$	$\frac{\log \pi + a/2 - \log \rho_1}{a}$	shell means converge to a positive constant; ordinary Cesaro means have a nonconstant dyadic phase profile
Root mean square ($q = 2$)	$2^{-1/2}$	$\frac{\log(2\pi)}{a}$	exact numerator identity $\int B_n^2 = 2^{-n}$; normalized shell RMS converges
Uniform envelope ($q = \infty$)	$\sqrt{3}/2$	$\frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a}$	global upper bound, sharp on $x = 2^n(2/3)$; not a two-sided envelope for every extremum

The ordering is

$$\begin{aligned} \kappa_\infty &= 2.359014879\ldots < \kappa_2 = 2.651496129\ldots \\ &< \kappa_1 = 2.748070014\ldots < \kappa_0 = 3.151496129\ldots. \end{aligned} \tag{1.3}$$

Larger q emphasizes rarer and larger values, hence gives a slower decay and a smaller power κ_q.

The coefficient $\log(2\pi)/\log 2$ appearing in the working draft is not an accidental near miss. It is *exactly* the root-mean-square power κ_2. It is not the arithmetic-mean power, the typical power, or the sharp uniform-envelope power. No factor $(\log x)^p$ can convert one of these powers into another, because their discrepancy is already a nonzero multiple of $\log x$ in the logarithm of the amplitude.

2 Audit map and final verdict

The four earliest reports and the first comparative audit established the analytic backbone. Their apparent disagreement is mostly a disagreement about which functional of $|\Phi|$ is being estimated. The following compact table records the verdict on that backbone before the detailed proofs; the full twelve-source concordance, including Reports 5–8, the second-wave audit, and the gentle guide, appears in Section 18.

Source	Main contribution	Final assessment
Wave 1, Document 1 [21]	Exact dyadic reduction, typical law, martingale CLT/LIL, and a sharp pointwise envelope.	Substantively correct on these layers. Its LIL constant $\pi/\sqrt{2}$ is the correct one; the argument is completed in Section 7.
Wave 1, Document 2 [22]	Sharp raw dyadic-annulus maxima, Lambert-W boundary layer, and explicit dyadic troughs.	Correct and strictly sharper than the other documents on annular maxima. The proof is reconstructed in Section 13.
Wave 1, Document 3 [23]	Typical, extremal, arithmetic-mean, and L^q interpretations; an analytic Cesaro gauge.	Correct except that it faithfully inherited the published LIL constant π, which is too large by $\sqrt{2}$.
Wave 1, Document 4 [24]	Most systematic transfer-operator treatment, exact RMS identity, sharp envelope, phase profile, and fixed-offset valley theorem.	Correct modulo its explicitly imported Ruelle–Perron–Frobenius block; it omits the fluctuation and Lambert boundary-layer layers rather than contradicting them.
Aistleitner–Hofer–Larcher [17]	Sharp metric study of the same lacunary products and a rigorous five-digit enclosure of the L^1 Perron root.	The order estimates and L^1 input are valid. The printed LIL constant contains a factor-$\sqrt{2}$ normalization error, located precisely in Section 7.3.
Comparative audit [29]	Reconciles the four readings, locates the LIL error, proves a stabilizing-gauge obstruction, and adds a logarithmic-mean repair.	Essentially correct. Its phrase "every elementary gauge" must be read as the explicitly proved stabilizing and bounded-distortion classes. It does not exclude every scale-dependent analytic gauge, and the later adaptive construction proves that such a gauge exists.

The final answer can therefore be stated compactly.

Theorem 2.1 (Decay dictionary). *Let E_κ be defined by* (1.2).

1. *For Lebesgue-a.e. $y \in (1,2)$, $|\Phi(2^n y)| = E_{\kappa_0}(2^n y)\exp(o(n))$ as $n \to \infty$, with asymptotic variance $\pi^2/4$ per dyadic level, the exact finite variance in Theorem 7.1, and the corrected LIL of Theorem 7.2.*

2. *The normalized shell L^q means converge to positive constants for every fixed $0 < q < \infty$, with exponent κ_q determined by the Perron root of* (5.5)*. In particular $q = 1$ answers the normalized dyadic-shell L^1 question and $q = 2$ gives $\kappa_2 = \log_2(2\pi)$ exactly.*

3. *There is $C < \infty$ such that $|\Phi(x)| \le C E_{\kappa_\infty}(x)$ for $x \ge 1$, and equality up to one fixed constant holds on the ray $x = 2^n(2/3)$.*

4. *The raw annular maxima have the additional Lambert-W suppression stated in Theorem 13.3; the local lobe maxima have both sharp-envelope subsequences and the fixed-offset dyadic valleys of Theorem 14.2.*

5. *The natural gauge $A_1 E_{\kappa_1}$ solves the two-sided Cesaro boundedness problem and the dyadic-endpoint limit. Its unrestricted endpoints retain a nonconstant phase profile, and every bounded-distortion shell correction fails. Nevertheless, the shell-adaptive analytic gauge of Theorem 10.3 solves the literal unrestricted ordinary Cesaro problem.*

6. *Without shell adaptation, the multiplicatively natural logarithmic mean of Theorem 11.1 converges for every endpoint.*

Every assertion in this summary is proved later or is explicitly marked as depending on a named classical analytic input.

Proof. Items 1–3 are Theorems 5.2, 7.2 and 12.3; item 4 is Theorems 13.3 and 14.2. The first part of item 5 is Theorems 9.5 and 9.6; the obstruction and existence statements are Theorems 9.9 and 10.3. Item 6 is Theorem 11.1. □

3 Exact products, zeros, and scaling laws

3.1 Entire convergence and three product forms

The compact convergence of (1.1) follows from

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4), \qquad \sum_{h\geq 0} \frac{|z|^2}{4^h} < \infty$$

uniformly for z in a compact set. Hence Φ is even, real on $\mathbb{R}$, entire, and $\Phi(0) = 1$.

Vieta's product $\operatorname{sinc} t = \prod_{j\geq 1} \cos(t/2^j)$ and Euler's sine product give two other useful forms:

$$\Phi(z) = \prod_{j=1}^{\infty} \cos^j\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \tag{3.1}$$

Indeed, in the cosine product the factor with denominator 2^j occurs for exactly j pairs of indices. In the Weierstrass product, the integer m is a zero of the sinc factor at level h precisely when $2^h \mid m$, so its total multiplicity is $1 + \nu_2(m)$.

The elementary one-step scaling law is

$$\Phi(2x) = \operatorname{sinc}(2\pi x)\Phi(x), \qquad \Phi(x) = \operatorname{sinc}(\pi x)\Phi(x/2). \tag{3.2}$$

3.2 The corrected reciprocal-Gamma and q-Pochhammer tail

The 2018 question contained a useful local expansion, but the proposed final formula had the wrong sign, the wrong geometric denominator, and inconsistent indices. We record the correct identity because it is an efficient evaluation formula and because its proof cleanly separates local analytic information from the large-x decay problem.

For $|q| < 1$, write

$$(u; q)_\infty := \prod_{j=0}^{\infty} (1 - uq^j),$$

and put

$$c_k := \frac{\zeta(2k) - 1}{k} = \frac{(2\pi)^{2k}|B_{2k}|/(2k)! - 2}{2k} > 0. \tag{3.3}$$

Theorem 3.1 (Corrected Gamma–Pochhammer factorization). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \prod_{n=0}^{\infty} \frac{1}{\Gamma(1+z/2^n)\Gamma(1-z/2^n)}. \tag{3.4}$$

For an integer $m \geq 0$, define the tail

$$R_m(z) := \prod_{n=m}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right).$$

If $|z| < 2^{m+1}$, then

$$R_m(z) = \left(\frac{z^2}{4^m}; \frac{1}{4}\right)_\infty \exp\left(-\sum_{k=1}^{\infty} \frac{c_k}{1-4^{-k}} \left(\frac{z}{2^m}\right)^{2k}\right), \tag{3.5}$$

and therefore

$$\Phi(z) = \left(\prod_{n=0}^{m-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right)\right) R_m(z). \tag{3.6}$$

The empty prefix for $m = 0$ is 1.

Proof. Euler's reflection formula and $\Gamma(1+w) = w\Gamma(w)$ give

$$\operatorname{sinc}(\pi w) = \frac{1}{\Gamma(1+w)\Gamma(1-w)},$$

so multiplication over $w = z/2^n$ proves (3.4); local uniform convergence follows from the same $1 + \mathcal{O}(4^{-n})$ estimate used for the sinc product.

To prove the tail formula, isolate the nearest zeros at $w = \pm 1$. The function $\operatorname{sinc}(\pi w)/(1-w^2)$ is holomorphic and zero-free in $|w| < 2$, equals 1 at 0, and its analytic logarithm has the absolutely convergent expansion

$$\operatorname{sinc}(\pi w) = (1-w^2) \exp\left(-\sum_{k=1}^{\infty} c_k w^{2k}\right), \qquad |w| < 2. \tag{3.7}$$

One may obtain the coefficients either from the Taylor series of $\log\Gamma(1 \pm w)$ after isolating their nearest poles, or directly from Euler's product; the identity for c_k in (3.3) is Euler's evaluation of $\zeta(2k)$.

Apply (3.7) with $w = z/2^n$ and multiply over $n \geq m$. Absolute and locally uniform convergence in $|z| < 2^{m+1}$ permits interchange of the two sums. The algebraic factors give

$$\prod_{n=m}^{\infty} \left(1 - \frac{z^2}{4^n}\right) = \left(\frac{z^2}{4^m}; \frac{1}{4}\right)_\infty,$$

while the analytic correction is

$$-\sum_{n=m}^{\infty} \sum_{k=1}^{\infty} c_k \frac{z^{2k}}{4^{nk}} = -\sum_{k=1}^{\infty} \frac{c_k}{1-4^{-k}} \left(\frac{z}{2^m}\right)^{2k}.$$

This is (3.5); splitting the original product at m gives (3.6). $\square$

Remark 3.2 (What the convergence disk does and does not say). The condition $|z| < 2^{m+1}$ is the convergence disk of the chosen logarithmic expansion, not a support statement for Φ. Outside it, formula (3.5) must be analytically continued or recomputed with a larger m; it emphatically does not say that the entire function vanishes there. Thus the free index m is a legitimate numerical truncation parameter. This factorization answers the local evaluation subproblem in the original post, but by itself does not determine the large-x Cesaro or extremal asymptotics.

The reciprocal-Gamma and Pochhammer identities are represented by exact declarations in the current Lean development, in modules `GeometricReciprocalGamma` and `RvachevPochhammerFactorization`. The exact tail-series refinement (3.5) is presently a human proof and a natural next formalization target.

3.3 Zeros and their multiplicities

Proposition 3.3 (Integer zero divisor). *The nonzero zeros of Φ are exactly the nonzero integers. The zero at $m \in \mathbb{Z} \setminus \{0\}$ has multiplicity*

$$d(m) = 1 + \nu_2(|m|).$$

Moreover, if $b \geq 1$ is an integer, $d = 1 + \nu_2(b)$, and $z \uparrow b$, then

$$|\Phi(z)| = b^{-d} \left|\Phi(b/2^d)\right| (b - z)^d \big(1 + \mathcal{O}(b - z)\big). \tag{3.8}$$

Proof. The divisor statement follows immediately from the last product in (3.1). For the local coefficient, the vanishing factors are those with $0 \leq h \leq \nu_2(b)$. At $z = b$ each has a simple zero and

$$\left| \frac{\mathrm{d}}{\mathrm{d}z} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \Big|_{z=b} \right| = \frac{1}{b}.$$

The product of the remaining factors is

$$\prod_{h=d}^{\infty} \left|\operatorname{sinc}\left(\frac{\pi b}{2^h}\right)\right| = \left|\Phi(b/2^d)\right|.$$

Multiplication gives (3.8). □

3.4 Exact dyadic renormalization

Let

$$T(x) = 2x \pmod 1, \qquad s(x) = |\sin(\pi x)|, \qquad x \in [0, 1],$$

and define the lacunary product

$$B_n(y) := \prod_{j=1}^{n} s(T^j y) = \prod_{j=1}^{n} \left|\sin(\pi 2^j y)\right|. \tag{3.9}$$

Theorem 3.4 (Exact dyadic factorization). *For every $n \geq 1$ and $y \in [1/2, 1]$,*

$$|\Phi(2^n y)| = |\Phi(y)|\, \pi^{-n} y^{-n} 2^{-n(n+1)/2} B_n(y). \tag{3.10}$$

For any real κ, set

$$W_\kappa(y) := |\Phi(y)|\, y^\kappa \exp\left(\frac{(\log y)^2}{2a}\right). \tag{3.11}$$

Then the following normalized identity is exact:

$$\frac{|\Phi(2^n y)|}{E_\kappa(2^n y)} = W_\kappa(y) \exp(n(\kappa a - \log \pi - a/2))\, B_n(y). \tag{3.12}$$

Proof. Iterating (3.2) gives

$$\Phi(2^n y) = \Phi(y)\prod_{j=1}^{n}\operatorname{sinc}(\pi 2^j y).$$

The product of the denominators is $\pi^n y^n 2^{1+\cdots+n} = \pi^n y^n 2^{n(n+1)/2}$, proving (3.10). Write $L = \log(2^n y) = na + \log y$ and divide (3.10) by $\exp(-L^2/(2a) - \kappa L)$. The terms $-n\log y$ cancel, leaving (3.12). □

Equation (3.12) is the central identity of this article. The entire deterministic decay has been extracted into E_κ. The bounded phase factor W_κ and the dynamical product B_n contain all remaining information.

3.5 The Thue–Morse polynomial

Let $\tau_k = (-1)^{s_2(k)}$, where $s_2(k)$ is the number of ones in the binary expansion of k. The finite product identity

$$P_n(x) := \prod_{j=0}^{n-1}(1 - \mathrm{e}^{2\pi \mathrm{i} 2^j x}) = \sum_{k=0}^{2^n-1}\tau_k \mathrm{e}^{2\pi \mathrm{i} k x} \tag{3.13}$$

follows by induction. Taking moduli yields

$$|P_n(x)| = 2^n \prod_{j=0}^{n-1} s(T^j x). \tag{3.14}$$

Thus the same products governing the decay of $\widehat{\mathrm{up}}$ are the normalized moduli of classical Thue–Morse trigonometric polynomials. This is the bridge to Gelfond exponents, Riesz products, and thermodynamic formalism; see [7, 15, 6].

Corollary 3.5 (Finite Thue–Morse sine expansion)**.** *For every integer $m \geq 0$ and every real $x \neq 0$,*

$$\prod_{h=0}^{m}\operatorname{sinc}\left(\frac{\pi x}{2^h}\right) = \frac{2^{m(m-1)/2}}{(\pi x)^{m+1}}\sum_{r=0}^{2^m-1}\tau_r \sin\left(\frac{\pi m}{2} + \frac{(2r+1)\pi x}{2^m}\right). \tag{3.15}$$

At $x = 0$ the identity holds by continuous extension, with value 1.

Proof. Put $\theta = \pi x/2^m$. Reversing the factors gives

$$\prod_{h=0}^{m}\sin\left(\frac{\pi x}{2^h}\right) = \prod_{j=0}^{m}\sin(2^j\theta).$$

Apply (3.13) with $m+1$ factors, multiply by the phase that converts $1 - \mathrm{e}^{2\mathrm{i}u}$ to $-2\mathrm{i}\mathrm{e}^{\mathrm{i}u}\sin u$, and pair the terms indexed by r and $2^{m+1} - 1 - r$. Binary complementation gives $\tau_{2^{m+1}-1-r} = (-1)^{m+1}\tau_r$; taking the appropriate real or imaginary part yields

$$\prod_{j=0}^{m}\sin(2^j\theta) = 2^{-m}\sum_{r=0}^{2^m-1}\tau_r\sin\left(\frac{\pi m}{2} + (2r+1)\theta\right).$$

Finally,

$$\prod_{h=0}^{m}\frac{\pi x}{2^h} = (\pi x)^{m+1}2^{-m(m+1)/2}.$$

Division proves (3.15); continuity handles $x = 0$. □

This is the cleaned form of the valid finite identity in the original 2018 post and is represented in Lean by `LacunaryProductToSum`.

Remark 3.6 (Where the Gaussian q-binomial analogy enters). The Thue–Morse identity and the finite q-binomial theorem

$$\prod_{j=0}^{n-1}(1+zq^j) = \sum_{r=0}^{n} q^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_q z^r$$

are two versions of the same subset-expansion principle. The Gaussian identity groups subsets by cardinality; the binary product $\prod_{j<n}(1-z^{2^j})$ instead assigns every subset its unique binary weight, leaving the coefficient $(-1)^{s_2(k)}$. This is why Gaussian-binomial arithmetic and Thue–Morse Fourier products meet elsewhere in the Fabius development even though the decay cocycle is most naturally written in binary-dynamical language.

3.6 A second exact scaling identity

The behavior just to the right of a large dyadic integer is exposed by a different exact identity.

Lemma 3.7 (Dyadic-boundary identity)**.** *Let $N = 2^{n-1}$ and $0 < z < N$. Then*

$$|\Phi(N+z)| = \frac{\left|\Phi(\frac{1}{2} + z/(2N))\right|}{|\Phi(z/(2N))|} \left(\frac{z}{N+z}\right)^n |\Phi(z)| \,. \tag{3.16}$$

Proof. For $0 \le h < n$, the number $N/2^h$ is an integer, so

$$\left|\sin(\pi(N+z)/2^h)\right| = \left|\sin(\pi z/2^h)\right|.$$

The quotient of the first n sinc factors is therefore $(z/(N+z))^n$ in absolute value. Reindexing the remaining tails gives

$$\prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi(N+z)}{2^h}\right) = \Phi\left(\frac{1}{2} + \frac{z}{2N}\right), \quad \prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) = \Phi\left(\frac{z}{2N}\right).$$

Combining the finite and infinite parts proves the identity. □

4 Exactly one extremum per unit interval

The Weierstrass product gives a short complete proof of the observed one-extremum property.

Theorem 4.1 (Strict logarithmic concavity between zeros)**.** *The function Φ is positive and strictly decreasing on $(0,1)$. For every integer $m \ge 1$, $|\Phi|$ has exactly one critical point $\xi_m \in (m, m+1)$. It is the strict maximum of $|\Phi|$ on that interval. Moreover, for every integer $m \ge 0$,*

$$\operatorname{sgn}\Phi(x) = (-1)^{s_2(m)}, \qquad m < x < m+1. \tag{4.1}$$

For $m \ge 1$ the signed extremum is therefore $\Phi(\xi_m) = (-1)^{s_2(m)} M_m$, where

$$M_m := \max_{m \le x \le m+1} |\Phi(x)| \,.$$

Proof. Away from the integer zeros, termwise logarithmic differentiation of (3.1) is locally uniform and gives

$$\frac{\mathrm{d}}{\mathrm{d}x} \log|\Phi(x)| = -2x \sum_{k=1}^{\infty} \frac{1+\nu_2(k)}{k^2 - x^2}, \tag{4.2}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\log|\Phi(x)| = -2\sum_{k=1}^{\infty}(1+\nu_2(k))\frac{k^2+x^2}{(k^2-x^2)^2} < 0. \tag{4.3}$$

For $0 < x < 1$, every denominator in (4.2) is positive, so $(\log\Phi)'(x) < 0$. This proves strict decrease on the first interval.

Now let $m \geq 1$. The logarithmic derivative is strictly decreasing on $(m, m+1)$. The pole contributed by $k = m$ forces it to $+\infty$ at the left endpoint, while the pole contributed by $k = m+1$ forces it to $-\infty$ at the right endpoint. It therefore has exactly one zero, and strict log-concavity makes that point the unique maximum of the modulus.

Starting from $\Phi(0) = 1$, crossing the zero at k changes the sign $1 + \nu_2(k)$ times. Therefore the sign on $(m, m+1)$ is

$$(-1)^{\sum_{k=1}^{m}(1+\nu_2(k))}.$$

Legendre's identity $\nu_2(m!) = m - s_2(m)$ gives

$$\sum_{k=1}^{m}(1+\nu_2(k)) = m + \nu_2(m!) = 2m - s_2(m),$$

whose parity is $s_2(m)$. This proves (4.1). □

Remark 4.2. The critical points do *not* in general converge to the midpoints of their unit intervals. In Section 14 we prove that, in every fixed interval offset from a large power of two, the critical point approaches the *right* endpoint at rate $1/n$.

4.1 A quantitative peak–area comparison

The unique height M_n and the area under the nth lobe differ by at most a polylogarithmic factor. This turns qualitative log-concavity into a reusable quantitative estimate.

Lemma 4.3 (Peak–integral comparison)**.** *There is an absolute constant C such that, for every integer $n \geq 2$,*

$$\int_n^{n+1}|\Phi(x)|\,\mathrm{d}x \leq M_n \leq C(1+\log n)^{5/2}\int_n^{n+1}|\Phi(x)|\,\mathrm{d}x. \tag{4.4}$$

In particular, the weaker exponent 3 stated in the comparative audit also holds.

Proof. The first inequality is immediate because the interval has length one. Put $F(x) = \log|\Phi(x)|$ on $(n, n+1)$, $d_0 = 1 + \nu_2(n)$, $d_1 = 1 + \nu_2(n+1)$, and write $\delta_0 = \xi_n - n$, $\delta_1 = n + 1 - \xi_n$. Isolating the two adjacent poles in (4.2) gives

$$F'(x) = \frac{d_0\theta_0(x)}{x-n} - \frac{d_1\theta_1(x)}{n+1-x} + R_n(x), \qquad \frac{4}{5} \leq \theta_i(x) \leq \frac{6}{5}, \tag{4.5}$$

for $n < x < n+1$. The remaining terms satisfy

$$|R_n(x)| \leq C_0(1+\log n)^2. \tag{4.6}$$

Indeed, for $k < n$ or $n+1 < k \leq 2n$ the kth summand is at most a constant times $(1+\log n)/|k-x|$; summing the harmonic series gives $\mathcal{O}((1+\log n)^2)$. For $k > 2n$, it is at most $Cn(1+\log k)/k^2$, whose tail is $\mathcal{O}(1+\log n)$. This proves (4.6).

At $x = \xi_n$ the derivative vanishes. If $\delta_0 \leq 1/2$, then $\delta_1 \geq 1/2$, and (4.5) gives $d_0/\delta_0 \leq C(1+\log n)^2$; since $d_0 \geq 1$, $\delta_0 \geq c(1+\log n)^{-2}$. Interchanging the endpoints handles $\delta_1 \leq 1/2$. Hence

$$\delta_* := \min(\delta_0, \delta_1) \geq c(1+\log n)^{-2}. \tag{4.7}$$

For $|x-\xi_n| \le \delta_*/2$, isolate the same two terms in (4.3). They contribute at most $C(1+\log n)\delta_*^{-2}$; the remaining inverse-square series is $\mathcal{O}(1+\log n)$. Thus

$$|F''(x)| \le \Lambda_n := C_1(1+\log n)^5. \tag{4.8}$$

Taylor's formula at the critical point gives $F(\xi_n+s) \ge F(\xi_n) - \Lambda_n s^2/2$. On the interval

$$|s| \le w_n := \min\{\delta_*/2, \Lambda_n^{-1/2}\} \ge c_1(1+\log n)^{-5/2},$$

we therefore have $|\Phi(\xi_n+s)| \ge \mathrm{e}^{-1/2} M_n$. Integration over this interval proves the second inequality in (4.4). □

The exact log-concavity and one-peak inputs are represented by exact declarations in the current Lean source; the explicit uniform bounds (4.6)–(4.8) remain a useful Lean target. Computations in the audits suggest that the actual ratio $M_n / \int_n^{n+1} |\Phi|$ is far smaller than this worst-case bound and that its shell aggregate may approach $1.92372\ldots$. No certified limiting constant is presently claimed. Likewise, a Gaussian limit for the distribution of the lobe heights is supported by the continuous martingale CLT and a regular-cell argument, but the required uniform cell-to-maximum transfer has not yet been formalized; it remains a stated research frontier, not a theorem of this volume.

5 Dyadic-shell means and the transfer operator

5.1 Normalized shell means

For $q \in (0,\infty)$ and a measurable H on the shell $D_n = [2^{n-1}, 2^n]$, define

$$\mathcal{M}_{q,n}(H) := \left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |H(x)|^q \,\mathrm{d}x\right)^{1/q} = \left(2\int_{1/2}^{1} |H(2^n y)|^q \,\mathrm{d}y\right)^{1/q}. \tag{5.1}$$

The endpoint conventions are

$$\mathcal{M}_{0,n}(H) := \exp\left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} \log|H(x)| \,\mathrm{d}x\right), \tag{5.2}$$

$$\mathcal{M}_{\infty,n}(H) := \sup_{2^{n-1}\le x\le 2^n} |H(x)|\,. \tag{5.3}$$

The logarithm in (5.2) is locally integrable despite the finitely many zeros.

After $y = (x+1)/2$, the numerator in (3.12) becomes

$$\prod_{j=0}^{n-1} s(T^j x), \qquad 0 \le x \le 1. \tag{5.4}$$

This product is generated by an explicit Perron–Frobenius operator.

5.2 The weighted doubling operator

For $q > 0$, define

$$(\mathcal{L}_q f)(x) := \frac{1}{2}\left[\sin^q\!\left(\frac{\pi x}{2}\right) f(x/2) + \cos^q\!\left(\frac{\pi x}{2}\right) f((x+1)/2)\right]. \tag{5.5}$$

The factor $1/2$ makes this the Lebesgue Perron operator with weight s^q. For every integrable f,

$$\int_0^1 f(x) \prod_{j=0}^{n-1} s(T^j x)^q \,\mathrm{d}x = \int_0^1 (\mathcal{L}_q^n f)(x)\,\mathrm{d}x. \tag{5.6}$$

We use the following standard Ruelle–Perron–Frobenius fact. The weights in (5.5) are nonnegative rather than strictly positive at the two endpoints, but the two-branch system is primitive, so the usual cone and Lasota–Yorke proof applies without alteration; see [3, Chs. 2–3] and the singular-potential discussion in [5, 6].

Proposition 5.1 (Perron root and spectral decomposition; classical RPF input)**.** *For every $q > 0$, $\mathcal{L}_q$ has a simple leading eigenvalue $\rho_q > 0$, a strictly positive eigenfunction h_q, and a positive eigenmeasure ν_q, which may be normalized by*

$$\mathcal{L}_q h_q = \rho_q h_q, \qquad \mathcal{L}_q^* \nu_q = \rho_q \nu_q, \qquad \nu_q(h_q) = 1.$$

On $C^\alpha[0,1]$, with $0 < \alpha \le \min(1,q)$ and norm $\|f\|_\infty + [f]_\alpha$, there is a spectral gap and

$$\rho_q^{-n} \mathcal{L}_q^n f = \nu_q(f) h_q + \mathcal{O}(\vartheta_q^n \|f\|_{C^\alpha}), \qquad 0 < \vartheta_q < 1. \tag{5.7}$$

Classical analytic input and proof architecture. Write the two inverse branches of doubling as $\tau_0(x) = x/2$ and $\tau_1(x) = (x+1)/2$, and their weights as $w_0(x) = \frac{1}{2}\sin^q(\pi x/2)$ and $w_1(x) = \frac{1}{2}\cos^q(\pi x/2)$. For $0 < \alpha \le \min(1,q)$ the weights belong to C^α, while both branches contract Holder seminorms by $2^{-\alpha}$. The usual Lasota–Yorke estimate therefore splits $\mathcal{L}_q$ into a strict seminorm contraction plus a compact $C^\alpha \hookrightarrow C^0$ part, giving quasi-compactness.

The only zeros of the weights occur at branch endpoints. On every compact subinterval of $(0,1)$ all admissible cylinder weights are positive, and topological exactness of doubling eventually sends every nonempty open interval across the whole unit interval. The positive cone is consequently irreducible and aperiodic; the Krein–Rutman theorem gives a positive leading eigenfunction and eigenmeasure, and the standard cone contraction excludes every other peripheral eigenvalue. Normalizing by $\nu_q(h_q) = 1$ yields the rank-one projector $f \mapsto \nu_q(f)h_q$; quasi-compactness gives the exponentially decaying remainder in (5.7). The test function used below is $W_{\kappa_q}((\cdot+1)/2)^q \in C^\alpha[0,1]$: its only endpoint zero has order q. Thus no unmentioned weighted space is needed. These are the hypotheses and conclusion of the Ruelle–Perron–Frobenius theorem cited above.

This operator-theoretic block is imported rather than presently reproved in Lean. All later uses cite exactly (5.7); the product algebra before and after it is independent of this input. □

5.3 The full finite-q decay spectrum

Set

$$\Lambda_q := \rho_q^{1/q}, \qquad \kappa_q := \frac{\log \pi + a/2 - \log \Lambda_q}{a}, \qquad q > 0. \tag{5.8}$$

Theorem 5.2 (Sharp finite-q shell asymptotics; conditional on Theorem 5.1)**.** *For every $q \in (0,\infty)$ there is a constant $A_q \in (0,\infty)$ such that*

$$\mathcal{M}_{q,n}\left(\frac{\Phi}{E_{\kappa_q}}\right) \longrightarrow A_q \qquad (n \to \infty). \tag{5.9}$$

More explicitly, with

$$\widetilde{W}_q(x) := W_{\kappa_q}\left(\frac{x+1}{2}\right)^q,$$

and the normalization in Theorem 5.1,

$$A_q^q = \left(\int_0^1 h_q(x)\,\mathrm{d}x\right) \nu_q(\widetilde{W}_q). \tag{5.10}$$

The number Λ_q is nondecreasing in q, and therefore κ_q is nonincreasing.

Proof. Raise (3.12) to the qth power, use (5.1), and substitute $x = 2y - 1$. This gives

$$\mathcal{M}_{q,n}\left(\frac{\Phi}{E_\kappa}\right)^q = \exp(qn(\kappa a - \log \pi - a/2)) \\ \times \int_0^1 W_\kappa\left(\frac{x+1}{2}\right)^q \prod_{j=0}^{n-1} s(T^j x)^q \, \mathrm{d}x.$$

By (5.6) and (5.7), the integral equals

$$\rho_q^n \left(\int_0^1 h_q\right) \nu_q(W_\kappa((\cdot\, + 1)/2)^q)\,(1 + o(1)).$$

The definition (5.8) is exactly the cancellation condition

$$q(\kappa_q a - \log \pi - a/2) + \log \rho_q = 0.$$

This proves (5.9) and (5.10).

For each n, the L^q norm of the product (5.4) on the probability space $[0, 1]$ is nondecreasing in q. Taking nth roots and then the limit proves that Λ_q is nondecreasing. Formula (5.8) then gives the opposite monotonicity of κ_q. □

Remark 5.3 (No logarithmic-power correction). For a fixed $q \in (0, \infty)$, replacing E_{κ_q} by $E_{\kappa_q}(\log x)^p$ multiplies the normalized shell mean by $(na)^{-p}(1 + o(1))$. It therefore tends to 0 when $p > 0$ and to $+\infty$ when $p < 0$. At the sharp power, the required logarithmic exponent is $p = 0$.

6 Geometric-mean and typical decay

The $q = 0$ endpoint is especially simple because Lebesgue measure is invariant under the doubling map and

$$\int_0^1 \log \sin(\pi x)\, \mathrm{d}x = -\log 2 = -a. \tag{6.1}$$

Theorem 6.1 (Exact shell geometric mean)**.** *Define*

$$\kappa_0 := \frac{3}{2} + \frac{\log \pi}{a} = 3.1514961294723187980\ldots. \tag{6.2}$$

Then

$$\mathcal{M}_{0,n}\left(\frac{\Phi}{E_{\kappa_0}}\right) = \exp\left(\int_0^1 \log W_{\kappa_0}\left(\frac{x+1}{2}\right) \mathrm{d}x\right) \tag{6.3}$$

for every $n \geq 1$. *In particular, the normalized shell geometric mean is not merely asymptotically constant; it is exactly independent of the shell.*

Proof. Take logarithms in (3.12), substitute $x = 2y - 1$, and average. The n logarithmic sine terms each have integral $-a$ by invariance and (6.1). For (6.2),

$$\kappa_0 a - \log \pi - a/2 = a,$$

so the linear contribution na cancels the integral $-na$ exactly. □

The same exponent governs almost every fixed dyadic phase.

Corollary 6.2 (Almost-everywhere dyadic-ray decay). *For Lebesgue-a.e.* $y \in [1/2, 1]$,

$$\log|\Phi(2^n y)| = -\frac{(\log(2^n y))^2}{2a} - \kappa_0 \log(2^n y) + o(n). \tag{6.4}$$

Equivalently,

$$|\Phi(2^n y)| = E_{\kappa_0}(2^n y)\mathrm{e}^{o(n)}.$$

Proof. The function $\log s$ is integrable and the doubling map is ergodic. Birkhoff's theorem therefore gives

$$\frac{1}{n}\log B_n(y) = \frac{1}{n}\sum_{j=1}^{n}\log s(T^j y) \longrightarrow -a$$

for almost every y. Substitute this in (3.12) with $\kappa = \kappa_0$. □

The $o(n)$ in (6.4) is genuinely dynamical; it need not converge to a constant. Thus even the "typical" statement is a logarithmic asymptotic, not a pointwise equivalent with one universal prefactor.

7 Central limit theorem and the corrected iterated-logarithm law

The almost-everywhere exponent records only the mean of the logarithmic cocycle. Its next order is probabilistic, but in this particular problem the variance and the martingale structure are completely explicit. Put

$$\psi(t) := \log|2\sin(\pi t)|, \qquad S_n(t) := \sum_{j=0}^{n-1}\psi(T^j t), \qquad Tt = 2t \pmod 1. \tag{7.1}$$

The exact centered logarithm in the alternative shell coordinates $x = 2^k(1+t)$ is

$$\log|\Phi(x)| = -\frac{(\log x)^2}{2a} - \kappa_0 \log x + \Psi(1+t) + S_{k+1}(t), \tag{7.2}$$

where the mantissa term is explicitly

$$\Psi(y) := \log W_{\kappa_0}(y/2), \qquad 1 < y < 2. \tag{7.3}$$

It is bounded on every compact subinterval of $(1,2)$. Thus all metric fluctuations are those of S_n.

7.1 Exact covariance and finite-n variance

The classical Fourier expansion

$$\psi(t) = -\sum_{m=1}^{\infty}\frac{\cos(2\pi m t)}{m} \quad \text{in } L^2(0,1) \tag{7.4}$$

contains the whole calculation.

Theorem 7.1 (Exact variance). *The function ψ has mean zero and, for every $n \geq 1$,*

$$\int_0^1 S_n(t)^2\,\mathrm{d}t = \frac{\pi^2}{4}n - \frac{\pi^2}{3}\big(1 - 2^{-n}\big). \tag{7.5}$$

Consequently the asymptotic variance per dyadic level is

$$\sigma^2 = \lim_{n\to\infty}\frac{1}{n}\int_0^1 S_n^2 = \frac{\pi^2}{4}. \tag{7.6}$$

Proof. The mean is zero by the classical log-sine integral. Parseval and $\int_0^1 \cos^2(2\pi mt)\,\mathrm{d}t = 1/2$ give

$$C_0 := \int_0^1 \psi(t)^2\,\mathrm{d}t = \frac{1}{2}\sum_{m\geq 1}\frac{1}{m^2} = \frac{\pi^2}{12}.$$

For $r \geq 1$, orthogonality leaves only frequencies $m = 2^r\ell$:

$$C_r := \int_0^1 \psi(t)\psi(T^r t)\,\mathrm{d}t = \frac{1}{2}\sum_{\ell\geq 1}\frac{1}{2^r\ell}\frac{1}{\ell} = \frac{\pi^2}{12\,2^r}.$$

Stationarity now yields

$$\int S_n^2 = nC_0 + 2\sum_{r=1}^{n-1}(n-r)C_r.$$

Since $\sum_{r=1}^{n-1}(n-r)2^{-r} = n - 2 + 2^{1-n}$, this is exactly (7.5). □

7.2 A one-line Gordin decomposition

Let $\mathcal{P}$ be the unweighted Perron operator of the doubling map,

$$(\mathcal{P}f)(x) = \frac{1}{2}\left(f(x/2) + f((x+1)/2)\right).$$

From (7.4), $\mathcal{P}\psi = \psi/2$. Therefore

$$d := 2\psi - \psi\circ T = \log|\tan(\pi t)|, \qquad \mathcal{P}d = 0, \tag{7.7}$$

and telescoping gives the exact identity

$$S_n = \sum_{j=0}^{n-1} d\circ T^j - \psi + \psi\circ T^n. \tag{7.8}$$

The sequence $(d\circ T^j)$ is a stationary ergodic reverse martingale-difference sequence for the decreasing filtration $(T^{-j}\mathcal{B})$. Moreover

$$\int_0^1 d(t)^2\,\mathrm{d}t = \frac{2}{\pi}\int_0^{\pi/2}\log^2(\tan u)\,\mathrm{d}u = \frac{\pi^2}{4}. \tag{7.9}$$

The final integral follows from the standard log-sine and log-cosine integrals; it also agrees with (7.6), as it must.

Theorem 7.2 (CLT and corrected LIL). *With respect to Lebesgue measure on $t \in (0,1)$,*

$$\frac{S_n}{\sqrt{n}} \implies N\left(0, \frac{\pi^2}{4}\right). \tag{7.10}$$

For Lebesgue-a.e. t,

$$\limsup_{n\to\infty}\frac{S_n(t)}{\sqrt{2n\log\log n}} = \frac{\pi}{2}, \qquad \liminf_{n\to\infty}\frac{S_n(t)}{\sqrt{2n\log\log n}} = -\frac{\pi}{2}. \tag{7.11}$$

Equivalently, with denominator $\sqrt{n\log\log n}$, the two constants are $\pm\pi/\sqrt{2}$.

Proof. The tail estimate $\mathrm{Leb}\{|d| > u\} = O(\mathrm{e}^{-u})$ shows that $d \in L^p(0,1)$ for every finite p. With $\mathcal{F}_j = T^{-j}\mathcal{B}$, the variables $d \circ T^j$ form a stationary reverse martingale-difference sequence. Moreover,

$$\frac{1}{n}\sum_{j=0}^{n-1}\mathbb{E}\big[(d\circ T^j)^2 \mid \mathcal{F}_{j+1}\big] = \frac{1}{n}\sum_{j=0}^{n-1}(\mathcal{P}d^2)\circ T^{j+1} \longrightarrow \int_0^1 d^2$$

almost surely by the ergodic theorem, and the conditional Lindeberg condition follows from $d \in L^2$. For clarity, the extra LIL hypotheses are not being hidden: the predictable quadratic variation is the displayed ergodic sum and hence is asymptotic to $n\int d^2$, while the exponential tail and Borel–Cantelli give $\max_{j\le n}|d\circ T^j| = O(\log n) = o(\sqrt{n/\log\log n})$ almost surely. Truncating at $\varepsilon\sqrt{n/\log\log n}$ and using the same tail bound gives the conditional Lindeberg summability required by Stout's martingale LIL. Thus the stationary martingale CLT and the martingale LIL apply to $\sum_{j<n} d\circ T^j$; see Stout [19] and the invariance-principle formulation of Heyde–Scott [20]. These two classical martingale theorems are the imported probabilistic input. The variance is (7.9). It remains only to remove the two coboundary terms in (7.8). The positive part of ψ is bounded by $\log 2$, while near the integers

$$\mathrm{Leb}\{\psi < -u\} = O(\mathrm{e}^{-u}).$$

Hence $\sum_n \mathrm{Leb}\{|\psi\circ T^n| > \varepsilon\sqrt{n\log\log n}\} < \infty$. Borel–Cantelli makes the boundary terms $o(\sqrt{n\log\log n})$ almost surely. Replacing the threshold by $\varepsilon\sqrt{n}$ also makes them $o(\sqrt{n})$ in probability, which completes the CLT. □

In terms of the Fourier transform, for a.e. fixed $y = 1+t \in (1,2)$ and $x_k = 2^k y$, (7.2) and (7.11) give

$$\limsup_{k\to\infty}\frac{\log|\Phi(x_k)| + (\log x_k)^2/(2a) + \kappa_0\log x_k - \Psi(y)}{\sqrt{2(k+1)\log\log(k+1)}} = \frac{\pi}{2}, \tag{7.12}$$

with the corresponding negative liminf. The multiplicative fluctuation is therefore

$$\exp\left(\pm\left(\frac{\pi}{\sqrt{2}} + o(1)\right)\sqrt{\frac{\log x}{\log 2}\,\log\log\log x}\right),$$

which is subpolynomial in x but larger than every fixed power of $\log x$.

7.3 The normalization error in the published theorem

Theorem 1.2 of Aistleitner–Hofer–Larcher [17] prints the upper and recurrent lower constants as π when the denominator is $\sqrt{n\log\log n}$. That value is incompatible with the exact variance (7.5). On p. 663, eq. (4.5), the paper's variance calculation prints the cosine-series diagonal without the factor $\int_0^1 \cos^2(2\pi m t)\,\mathrm{d}t = 1/2$, thereby doubling σ^2 from $\pi^2/4$ to $\pi^2/2$. Restoring that factor gives $\sigma = \pi/2$ and therefore the constant $\sqrt{2}\sigma = \pi/\sqrt{2}$ in the $\sqrt{n\log\log n}$ normalization. This correction changes no exponent or qualitative conclusion in that paper, but it matters for the sharp constant.

8 The exact RMS power

For $q = 2$, the transfer operator simplifies dramatically:

$$\mathcal{L}_2\mathbf{1} = \frac{1}{2}\left(\sin^2\frac{\pi x}{2} + \cos^2\frac{\pi x}{2}\right) = \frac{1}{2}\mathbf{1}. \tag{8.1}$$

Its operator norm on $C[0,1]$ is at most $1/2$, so

$$\rho_2 = \frac{1}{2}, \qquad \Lambda_2 = \frac{1}{\sqrt{2}}. \tag{8.2}$$

There is also a completely finite Fourier proof. Parseval applied to (3.13) gives

$$\int_0^1 |P_n(x)|^2 \,\mathrm{d}x = 2^n,$$

while (3.14) gives

$$\int_0^1 \prod_{j=0}^{n-1} s(T^j x)^2 \,\mathrm{d}x = 2^{-n}. \tag{8.3}$$

Corollary 8.1 (RMS decay). *The sharp root-mean-square exponent is*

$$\boxed{\kappa_2 = \frac{\log(2\pi)}{\log 2} = 2.6514961294723187980\ldots.} \tag{8.4}$$

There is a constant $A_2 \in (0,\infty)$ such that

$$\mathcal{M}_{2,n}\left(\frac{\Phi}{E_{\kappa_2}}\right) \longrightarrow A_2. \tag{8.5}$$

No power of $\log x$ belongs in the leading RMS normalization.

Proof. Equations (8.1)–(8.2) give $\Lambda_2 = 2^{-1/2}$. Substitution in (5.8) gives

$$\kappa_2 = \frac{\log \pi + a/2 + (a/2)}{a} = \frac{\log(2\pi)}{\log 2}.$$

The convergence and positivity of A_2 are the $q = 2$ specialization of Theorem 5.2. Finally Theorem 5.3 forces logarithmic exponent zero. The independent finite identity (8.3) checks the spectral growth $\rho_2 = 1/2$ without any numerical approximation. □

Remark 8.2 (What the draft computed). The power $x^{-\log(2\pi)/\log 2}$ in [14] is precisely $x^{-\kappa_2}$. Thus the draft identified the quadratic-mean scale, albeit without specifying that norm. Its claimed pointwise equivalent and its proposed logarithmic correction do not follow and are incompatible with the exact shell identity (8.3), the integer zeros, and the exceptional extrema proved later.

8.1 Two exact finite-grid identities

For clarity in this subsection write

$$S_n(t) := \prod_{j=0}^{n-1} \left|\sin(\pi 2^j t)\right|.$$

Proposition 8.3 (Roots of unity and dyadic Parseval). *For every integer $n \ge 1$,*

$$\prod_{r=1}^{2^n-1} 2\sin\frac{\pi r}{2^n} = 2^n, \tag{8.6}$$

$$\sum_{m=0}^{2^n-1} S_n\left(\frac{m}{2^n}\right)^2 = 1. \tag{8.7}$$

Proof. Let $N = 2^n$ and $\omega = \mathrm{e}^{2\pi\mathrm{i}/N}$. Differentiating $z^N - 1 = (z-1)\prod_{r=1}^{N-1}(z-\omega^r)$ at $z = 1$ gives

$$N = \prod_{r=1}^{N-1}(1-\omega^r).$$

Taking absolute values and using $|1-\mathrm{e}^{2\pi\mathrm{i}r/N}| = 2\sin(\pi r/N)$ proves (8.6).

For the second identity, evaluate the Thue–Morse polynomial (3.13) on the Nth roots of unity. Discrete Parseval gives

$$\sum_{m=0}^{N-1} |P_n(m/N)|^2 = N \sum_{r=0}^{N-1} |\tau_r|^2 = N^2.$$

Since (3.14) says $|P_n(t)| = 2^n S_n(t) = N S_n(t)$, division by N^2 yields (8.7). □

Both statements are represented in the current Lean source as `SineProductRootsUnity` and `DiscreteLacunaryParseval`. Together with (8.3), they show that the continuous and dyadic-grid quadratic means of the numerator are exactly 2^{-n} at every level.

8.2 Exact transfer identities without a spectral overclaim

Proposition 8.4 (Explicit eigenfunctions of $\mathcal{L}_2$). *The quadratic transfer operator satisfies*

$$\mathcal{L}_2 \mathbf{1} = \tfrac{1}{2}\mathbf{1}, \tag{8.8}$$

$$\mathcal{L}_2[\sin(2\pi x)] = -\tfrac{1}{4}\sin(2\pi x), \tag{8.9}$$

$$\mathcal{L}_2[1+3\cos(2\pi x)] = -\tfrac{1}{4}[1+3\cos(2\pi x)]. \tag{8.10}$$

For $e_m(x) = \mathrm{e}^{2\pi\mathrm{i}mx}$ *it also obeys*

$$\mathcal{L}_2 e_{2m} = \tfrac{1}{2} e_m. \tag{8.11}$$

Proof. On both inverse branches, $e_{2m}(x/2) = e_m(x)$ and $e_{2m}((x+1)/2) = e_m(x)$; hence (8.11) follows from $\sin^2 + \cos^2 = 1$, and $m = 0$ gives (8.8). For frequency one,

$$\mathcal{L}_2 e_1 = \tfrac{1}{2}\mathrm{e}^{\pi\mathrm{i}x}\left(\sin^2\frac{\pi x}{2} - \cos^2\frac{\pi x}{2}\right) = -\tfrac{1}{4}(e_1 + \mathbf{1}).$$

The conjugate identity holds for e_{-1}. Taking their imaginary difference gives (8.9); combining their real sum with (8.8) gives (8.10). □

These pointwise identities are formalized in `RMSTransferEigenfunctions`. They prove that $-1/4$ is an eigenvalue, but they do *not* prove that it is the second eigenvalue or that all remaining spectrum has smaller modulus. Earlier reports inferred an alternating $(-1/2)^n$ relative RMS convergence rate from collocation; that rate remains numerical evidence until a certified spectral bound on a specified function space is supplied.

8.3 Plancherel, half-integer sampling, and total energy

Proposition 8.5 (Total Fourier energy). *One has the exact identities*

$$\int_{\mathbb{R}} |\Phi(x)|^2 \,\mathrm{d}x = \int_{\mathbb{R}} \mathrm{up}(t)^2\,\mathrm{d}t = \frac{1}{2} + \sum_{m=0}^{\infty} \Phi\left(m+\frac{1}{2}\right)^2. \tag{8.12}$$

Numerical quadrature and the rapidly convergent series both give $0.80887353596797056293124298\ldots$*; this decimal is computational evidence, not an interval certificate.*

Proof. The first equality is Plancherel, using that $\Phi = \widehat{\mathrm{up}}$ is real on the real axis. Expand the function up, supported in $[-1,1]$, in its length-two Fourier series. Its kth coefficient is $\frac{1}{2}\widehat{\mathrm{up}}(k/2) = \frac{1}{2}\Phi(k/2)$, so Parseval on $[-1,1]$ gives

$$\int_{-1}^{1} \mathrm{up}(t)^2\,\mathrm{d}t = \frac{1}{2}\sum_{k\in\mathbb{Z}} \Phi(k/2)^2.$$

The term $k = 0$ contributes $1/2$. Every nonzero even term vanishes because Φ is zero at every nonzero integer, and the positive and negative odd terms pair by evenness. This is exactly (8.12). Super-polynomial decay makes the series absolutely and rapidly convergent. □

The Plancherel/sinc-square part of this result is represented in Lean by the declaration

`FabiusSquareEnergyFourier`.

The half-integer sampling identity above is a concise human-readable target for completing that formalized energy layer.

9 Arithmetic-mean decay and the Cesaro question

9.1 The Perron root ρ_1

For $q = 1$,

$$(\mathcal{L}_1 f)(x) = \frac{1}{2}\left[\sin\frac{\pi x}{2}\, f(x/2) + \cos\frac{\pi x}{2}\, f((x+1)/2)\right]. \tag{9.1}$$

There appears to be no elementary closed form for its Perron root. It is nevertheless an exact, canonical constant, and it is easy to compute stably:

$$\rho_1 = 0.661322602060565\ldots. \tag{9.2}$$

The corresponding power is

$$\boxed{\kappa_1 = \frac{\log\pi + a/2 - \log\rho_1}{a} = 2.7480700148713326739\ldots.} \tag{9.3}$$

The decimal in (9.2) is numerical, but the following elementary Collatz–Wielandt certificate provides a rigorous enclosure.

Proposition 9.1 (A rational enclosure for ρ_1). *One has*

$$0.66126807 < \rho_1 < 0.66134891, \tag{9.4}$$

and consequently

$$2.74801262 < \kappa_1 < 2.74818899. \tag{9.5}$$

Proof. For a positive continuous test function f, the Collatz–Wielandt inequalities give

$$\inf_{x\in[0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)} \le \rho_1 \le \sup_{x\in[0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)}.$$

Take

$$f(x) = F(\sin\pi x), \qquad F(u) = 1 + \frac{376189}{10^6}u - \frac{69093}{10^6}u^2 + \frac{15483}{10^6}u^3.$$

This function is positive on $[0,1]$. Put $t = \tan(\pi x/4) \in [0,1]$, $u = 2t/(1+t^2)$, and $v = (1-t^2)/(1+t^2)$. Then

$$\frac{\mathcal{L}_1 f(x)}{f(x)} = \frac{uF(u) + vF(v)}{2F(2uv)},$$

a rational function of t. Exact Sturm-sequence checks show that the numerators of its differences from the two rational endpoints in (9.4) are positive and have no zeros on $[0,1]$. A compact exact-arithmetic verification is included in Section A. □

Remark 9.2 (Published enclosure). Aistleitner–Hofer–Larcher prove the sharper rigorous interval

$$0.66130 < \rho_1 < 0.66135, \tag{9.6}$$

by the Fouvry–Mauduit transfer recursion. It implies $2.74801024 < \kappa_1 < 2.74811933$. The wider interval above is retained because its certificate is short, exact, and executable from the appendix alone.

Remark 9.3 (Separate Lean certificate). The current Lean source contains a different, wider Bernstein-certificate bracket for the growth root defined from $\sup_x \mathcal{L}_1^n \mathbf{1}(x)$:

$$0.66126798 \le \rho \le 0.66134921, \qquad 2.7480118 \le \kappa(\rho) \le 2.7481893.$$

The exact declarations are `perron_root_enclosure` and `kappa_one_enclosure`. They do not formalize the sharper strict Sturm bracket (9.4). Moreover, identifying this growth root with the simple RPF eigenvalue ρ_1 uses the imported spectral theorem in Theorem 5.1; source-level existence and an enclosure should not be conflated with that analytic identification.

9.2 Sharp dyadic-shell mean

By Theorem 5.2, there is $A_1 > 0$ such that

$$2\int_{1/2}^{1} \frac{|\Phi(2^n y)|}{E_{\kappa_1}(2^n y)}\,\mathrm{d}y \longrightarrow A_1. \tag{9.7}$$

Therefore the choice

$$g_1(x) := A_1 E_{\kappa_1}(x) \tag{9.8}$$

makes the mean of $|\Phi|/g_1$ on each dyadic shell tend to one. Chebyshev collocation of the transfer operator, followed by the spectral formula (5.10), gives

$$A_1 = 0.091266124131588\ldots. \tag{9.9}$$

The decimal is numerical; the exact definition is the positive spectral expression (5.10). This gives the precise answer to the natural shell version of the question in [12].

Proposition 9.4 (Regularity of the arithmetic-mean gauge)**.** *The function $g_1(x) = A_1 E_{\kappa_1}(x)$ is positive and real-analytic on $(0,\infty)$ and strictly decreasing for $x \ge 1$. For every fixed integer $m \ge 0$ there is X_m such that*

$$(-1)^m g_1^{(m)}(x) > 0, \qquad x \ge X_m. \tag{9.10}$$

Thus it has every one of the eventual regularity properties requested in the original question, although the threshold may depend on the derivative order.

Proof. Put $u = \log x$ and $G(u) = g_1(\mathrm{e}^u)$. Then

$$G(u) = A_1 \exp\left(-\frac{u^2}{2a} - \kappa_1 u\right), \qquad x^m \frac{\mathrm{d}^m}{\mathrm{d}x^m} = D_u(D_u - 1)\cdots(D_u - m + 1).$$

Writing $p(u) = u/a + \kappa_1$, repeated differentiation gives

$$(-1)^m \frac{x^m g_1^{(m)}(x)}{g_1(x)} = p(u)^m + O_m(p(u)^{m-1}) \qquad (u \to \infty).$$

The right-hand side is positive for all sufficiently large u. The case $m = 1$ also follows directly from $g_1'(x)/g_1(x) = -(\log x/a + \kappa_1)/x < 0$ for $x \ge 1$. □

It also answers the weaker full-ray boundedness question.

Corollary 9.5 (Two-sided Cesaro boundedness)**.** *For every fixed $t_0 > 0$, there are constants $0 < c < C < \infty$ such that, for all sufficiently large t,*

$$c < \frac{1}{t - t_0}\int_{t_0}^{t} \frac{|\Phi(x)|}{E_{\kappa_1}(x)}\,\mathrm{d}x < C. \tag{9.11}$$

After multiplying E_{κ_1} by A_1, the same average tends to one along the dyadic endpoints $t = 2^n$.

Proof. Equation (9.7) says that the integral over the shell $[2^{n-1}, 2^n]$ is asymptotic to $A_1 2^{n-1}$. Summing the geometric sequence of completed shells proves the dyadic-endpoint limit. If t lies inside the next shell, the completed shells already contribute a positive constant times t, while the entire current shell contributes at most another constant times t. This gives (9.11). □

9.3 Why the ordinary Cesaro limit has a dyadic phase

The full limit in the stronger condition of [12] does not follow from the shell limit. In fact, the natural normalization (9.8) has a nonconstant phase profile. This can be stated exactly in terms of the eigenmeasure of $\mathcal{L}_1$.

Normalize h_1, ν_1 as in Theorem 5.1, and put

$$f_1(x) := W_{\kappa_1}\left(\frac{x+1}{2}\right), \qquad \mathscr{F}(z) := \frac{\nu_1(f_1 \mathbf{1}_{[0,z]})}{\nu_1(f_1)}, \quad 0 \le z \le 1.$$

The measure $f_1 \, \mathrm{d}\nu_1$ is non-atomic, so $\mathscr{F}$ is continuous. Here is a short verification. The endpoint atoms of ν_1 vanish because the corresponding transfer weights vanish. For $z \in (0,1) \setminus \{1/2\}$, the eigenmeasure equation gives

$$\rho_1 \nu_1(\{z\}) = \frac{1}{2} s(z) \nu_1(\{Tz\}).$$

(The possible atom at $1/2$ is fed only by the already vanishing endpoint atoms.) Hence any remaining atom would generate forward atoms whose masses grow by at least the factor $2\rho_1 > 1$ at every step. A periodic orbit gives an immediate contradiction, and a nonperiodic orbit would produce masses exceeding the finite total mass. Thus ν_1, and therefore $f_1 \, \mathrm{d}\nu_1$, has no atoms.

Theorem 9.6 (Dyadic phase limit). *For every $y \in [1/2, 1]$ and every fixed $t_0 > 0$,*

$$\lim_{n \to \infty} \frac{1}{2^n y - t_0} \int_{t_0}^{2^n y} \frac{|\Phi(x)|}{A_1 E_{\kappa_1}(x)} \, \mathrm{d}x = \mathscr{P}(y) := \frac{1 + \mathscr{F}(2y-1)}{2y}. \tag{9.12}$$

The continuous function $\mathscr{P}$ satisfies $\mathscr{P}(1/2) = \mathscr{P}(1) = 1$ but is not identically one. Consequently the ordinary Cesaro limit for the natural elementary scale $A_1 E_{\kappa_1}$ does not exist.

Proof. The spectral decomposition gives, for $z \in [0,1]$,

$$\rho_1^{-n} \int_0^z f_1(x) \prod_{j=0}^{n-1} s(T^j x) \, \mathrm{d}x \longrightarrow \left(\int_0^1 h_1\right) \nu_1(f_1 \mathbf{1}_{[0,z]}).$$

Approximation from above and below by Holder functions justifies the indicator; the leading conformal measure has no atoms. Returning to the shell variable, the partial integral from 2^{n-1} to $2^n y$ is asymptotic to

$$2^{n-1} A_1 \mathscr{F}(2y-1).$$

The sum of all preceding complete shells is asymptotic to $2^{n-1} A_1$. Division by $2^n y A_1$ gives (9.12).

It remains to prove that $\mathscr{F}$ is not the identity distribution. We first note that ν_1 is singular with respect to Lebesgue measure. Otherwise its nonzero absolutely continuous part would yield an L^1 density $r \ge 0$ satisfying

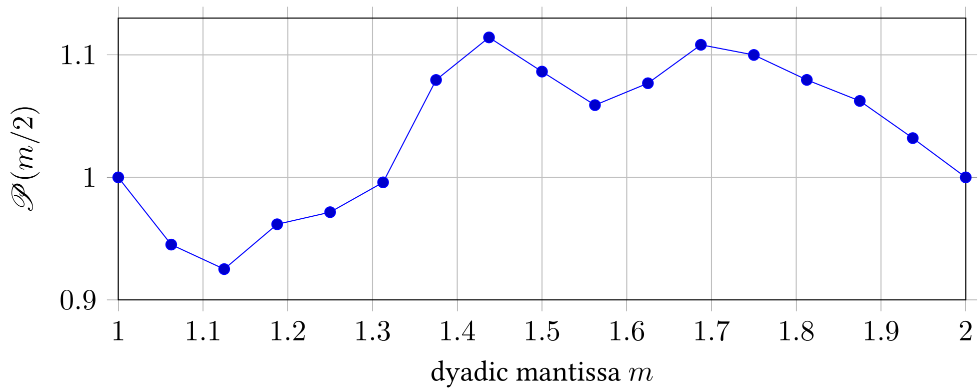
$$\rho_1 r(x) = s(x) r(Tx) \quad \text{for a.e. } x. \tag{9.13}$$

On the positive invariant support of r, iteration gives

$$r(T^n x) = \frac{\rho_1^n r(x)}{\prod_{j=0}^{n-1} s(T^j x)}.$$

For Lebesgue-a.e. x, Birkhoff's theorem makes the logarithmic growth rate of the right side $\log(2\rho_1) > 0$. On the other hand, for any $r \in L^1$ and any $\epsilon > 0$, Markov's inequality and Borel–Cantelli imply $r(T^n x) \le \mathrm{e}^{\epsilon n}$ eventually for almost every x. Choosing $0 < \epsilon < \log(2\rho_1)$ is a contradiction. Thus ν_1 is singular, and multiplying it by the continuous function f_1 leaves it singular. Hence its normalized distribution $\mathscr{F}$ cannot equal z, the distribution function of Lebesgue measure. But $\mathscr{P} \equiv 1$ would force $1 + \mathscr{F}(2y-1) = 2y$, or $\mathscr{F}(z) = z$. Therefore $\mathscr{P}$ is nonconstant. □

For orientation, the following finite-level computation displays the nonconstant profile in terms of the shell mantissa $m = 2y \in [1, 2]$. It uses the normalized density $f_1(z) \prod_{j<18} s(T^j z)$ on a dyadic midpoint grid; the theorem, not the plot, supplies the limiting statement.



Remark 9.7. A slowly varying correction, including every power of $\log x$, cannot remove this phase. For a slowly varying L, the ratio $L(2^n y)/L(2^n)$ tends uniformly to one for $y \in [1/2, 1]$, so the profile (9.12) survives. An exact all-phase normalizer would have to encode the singular conformal distribution, not merely add a smooth logarithmic factor.

9.4 How far the unrestricted Cesaro obstruction extends

The nonconstant profile in Theorem 9.6 is not an artifact of having omitted a power of $\log x$. It survives every gauge whose relative shape settles down inside successive dyadic shells.

Theorem 9.8 (Impossibility in the stabilizing class)**.** *Let $g : (X_0, \infty) \to (0, \infty)$ be a gauge. Suppose there are constants $c_k > 0$ and a continuous $u : [0, 1] \to (0, \infty)$ such that*

$$\frac{g(2^k(1+z))}{c_k E_{\kappa_1}(2^k(1+z))} \longrightarrow u(z) \quad \textit{uniformly for } 0 \le z \le 1. \tag{9.14}$$

Then the unrestricted limit

$$\frac{1}{t - X_0} \int_{X_0}^{t} \frac{|\Phi(x)|}{g(x)} \, dx \longrightarrow 1 \tag{9.15}$$

cannot hold.

Proof. Assume (9.15). If $F(t)$ denotes the numerator integral, then for every fixed $0 \le z_1 < z_2 \le 1$,

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} \longrightarrow z_2 - z_1. \tag{9.16}$$

The exact factorization in the coordinates $x = 2^k(1+z)$ gives

$$\frac{|\Phi(2^k(1+z))|}{E_{\kappa_1}(2^k(1+z))} = \rho_1^{-(k+1)} W(z) \prod_{j=0}^{k} |\sin(\pi 2^j z)|,$$

where W is continuous and positive on $[0, 1)$. Combining this identity with (9.14) and the Ruelle–Perron–Frobenius limit shows that every subsequential shell limit in (9.16) is a constant multiple of

$$\nu_1\left(\frac{W}{u} \mathbf{1}_{[z_1, z_2]}\right),$$

where ν_1 is the eigenmeasure of $\mathcal{L}_1^*$. The full-shell case forces the multiplicative constant to be finite and nonzero. Equation (9.16) would therefore make the measure $(W/u)\nu_1$ proportional to Lebesgue measure. Since W/u is continuous and positive on $[0,1)$ and ν_1 has no endpoint atoms, this would imply $\nu_1 \ll \mathrm{Leb}$.

That is impossible. If $\mathrm{d}\nu_1 = r\,\mathrm{d}x$ had a nonzero absolutely continuous part, the eigenmeasure equation would give

$$|\sin(\pi z)|r(Tz) = \rho_1 r(z) \quad \text{a.e.} \tag{9.17}$$

On the positive support of r, iteration and Birkhoff's theorem yield

$$r(T^n z) = r(z)\frac{\rho_1^n}{\prod_{j=0}^{n-1}|\sin(\pi T^j z)|} = r(z)\exp\big(n(\log(2\rho_1) + o(1))\big).$$

Here $2\rho_1 > 1$; for example, Cauchy–Schwarz together with the exact L^2 identity and the Gelfond upper bound gives $\rho_1 \geq 1/\sqrt{3} > 1/2$. On the other hand, invariance of Lebesgue measure, Markov's inequality, and Borel–Cantelli imply $r(T^n z) \leq \mathrm{e}^{\varepsilon n}$ eventually for a.e. z, for every $\varepsilon > 0$. Choosing $\varepsilon < \log(2\rho_1)$ is a contradiction. Thus ν_1 is singular and (9.15) cannot hold. □

The hypothesis covers all gauges of the form $E_{\kappa_1}(x)L(x)p(\{\log_2 x\})$, where L is regularly varying (slow variation is the most relevant case) and p is positive, continuous, and periodic. It also covers the usual log-exp-algebraic or Hardy-field candidates of the correct rough size whenever their relative logarithmic derivative has a finite limit, because the mean-value theorem then gives a uniform limiting shell shape. It does not, by itself, rule out a deliberately scale-dependent analytic gauge whose within-shell shape keeps changing and attempts to follow finer and finer dyadic cylinders. The next theorem removes the assumption that a limiting shell shape exists.

In shell coordinates put

$$D_k(z) := \frac{\left|\Phi(2^k(1+z))\right|}{g_1(2^k(1+z))}, \qquad \mathrm{d}\mu_k(z) := D_k(z)\,\mathrm{d}z, \qquad 0 \leq z \leq 1. \tag{9.18}$$

The RPF limit used in Theorem 9.6 says that $\mu_k \Rightarrow \mu$, $\mu_k([0,1]) \to 1$, where μ is an atomless, full-support singular probability measure. Full support follows because every open interval contains a dyadic cylinder and the positive interior transfer weights propagate positive eigenmeasure mass into every cylinder.

Theorem 9.9 (Bounded-distortion obstruction). *Let $X_0 > 0$ and let $g : [X_0,\infty) \to (0,\infty)$ be measurable. Suppose there are constants $0 < m < M < \infty$ and numbers $c_k > 0$ such that, for every large k and every $z \in [0,1]$,*

$$m \leq \frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \leq M. \tag{9.19}$$

Then g cannot satisfy the unrestricted limit (9.15).

Proof. Put $a_k := \mu_k([0,1])$ and $\bar{\mu}_k := a_k^{-1}\mu_k$. Then $a_k \to 1$ and $\bar{\mu}_k \Rightarrow \mu$. After the irrelevant scalar c_k is cancelled, the probability measure induced on the kth shell has the form

$$\mathrm{d}\eta_k(z) = \frac{q_k(z)\,\mathrm{d}\mu_k(z)}{\int_0^1 q_k\,\mathrm{d}\mu_k}, \qquad M^{-1} \leq q_k \leq m^{-1}.$$

Since $\int q_k\,\mathrm{d}\mu_k \geq a_k/M$, we have the correctly normalized domination $\eta_k \leq (M/m)\bar{\mu}_k$: for every continuous $\varphi \geq 0$,

$$\int \varphi\,\mathrm{d}\eta_k \leq \frac{M}{m}\int \varphi\,\mathrm{d}\bar{\mu}_k.$$

Any weak limit η is therefore dominated by $(M/m)\mu$ and is singular. But (9.15), evaluated at 2^k and $2^k(1+z)$ and subtracted, forces the partial shell masses to converge uniformly to z; hence η_k must converge to Lebesgue measure. This contradiction proves the theorem. □

Corollary 9.10 (Necessary shell microstructure)**.** *Every gauge satisfying the unrestricted ordinary Cesaro limit must have unbounded within-shell distortion relative to g_1, even after an arbitrary scalar renormalization on each shell.*

Proof. Otherwise some m, M and shell scalars c_k would satisfy (9.19), contradicting Theorem 9.9. □

The word "elementary" must not be substituted for the hypotheses above. For instance, if $2c\varepsilon < 1/a$, then

$$g_{\varepsilon,c}(x) := g_1(x)\exp\bigl(\varepsilon\sin(c(\log x)^2)\bigr)$$

is elementary, positive, real-analytic, and eventually decreasing, but its shell shape need not stabilize. It still fails by Theorem 9.9, because the multiplier is uniformly bounded above and below. Thus no implication "Hardy-field, hence elementary" is valid here.

10 A shell-adaptive analytic gauge with the exact unrestricted limit

The bounded-distortion theorem also tells us how an exact gauge must look: it must resolve progressively finer portions of the singular shell measure. The deterministic baseline is so steep that such a slowly increasing amount of microstructure can be inserted while the total gauge remains decreasing. This section gives the construction in full.

10.1 Smooth asymptotic flattening

Lemma 10.1 (Smooth flattening of weakly convergent measures)**.** *Let μ_k be finite positive Borel measures on $[0,1]$ converging weakly to an atomless, fully supported probability measure μ, with $\mu_k([0,1]) \to 1$. There are functions $r_k \in C^\infty([0,1])$, $r_k > 0$, such that*

1. *$r_k \equiv 1$ in neighborhoods of 0 and 1;*
2. $$\sup_{0\le z\le 1}\left|\int_0^z r_k\,\mathrm{d}\mu_k - z\right| \longrightarrow 0; \tag{10.1}$$
3. $$\|\log r_k\|_\infty = o(k), \qquad \bigl\|(\log r_k)'\bigr\|_\infty = o(k). \tag{10.2}$$

Proof. Choose finite partitions $0 = x_{j,0} < x_{j,1} < \cdots < x_{j,N_j} = 1$ whose meshes ε_j tend to zero. Because μ is atomless, the boundary points may be taken to be continuity points; because it has full support, every cell $I_{j,i} = [x_{j,i-1}, x_{j,i}]$ has positive mass. For fixed j, weak convergence gives $\mu_k(I_{j,i}) \to \mu(I_{j,i})$ simultaneously over the finite partition. Define the cell equalizer

$$c_{k,j,i} := \frac{|I_{j,i}|}{\mu_k(I_{j,i})}.$$

For fixed j, the finitely many ratios $c_{k,j,i}$ converge to positive finite limits. Choose B_j large enough that $|\log c_{k,j,i}| \le B_j/2$ for every cell and all large k. Now let U_j be a finite union of sufficiently small intervals around all partition boundaries, including 0 and 1, such that $\mu(\partial U_j) = 0$ and $\mu(U_j) + |U_j|$ is smaller than $\varepsilon_j e^{-B_j}/20$. Weak convergence then gives $\mu_k(U_j) \to \mu(U_j)$. Set $\ell_{k,j} = \log c_{k,j,i}$ on each cell outside U_j, join these finitely many constants smoothly inside U_j, and arrange $\ell_{k,j} = 0$ near the endpoints. The interpolation can be kept between its neighboring values and chosen with bounds

$$\|\ell_{k,j}\|_\infty \le B_j, \qquad \bigl\|\ell'_{k,j}\bigr\|_\infty \le C_j.$$

Put $r_{k,j} = \exp \ell_{k,j}$. On a cell $I = I_{j,i}$ the definition $c_{k,j,i}\mu_k(I) = |I|$ gives the exact error identity

$$\int_I r_{k,j}\,\mathrm{d}\mu_k - |I| = \int_{I\cap U_j} (r_{k,j} - c_{k,j,i})\,\mathrm{d}\mu_k.$$

Thus the total error over any union of complete cells is at most $2e^{B_j}\mu_k(U_j)$. A cumulative interval $[0,z]$ has, in addition, at most one partially traversed cell. Its portion outside U_j has weighted mass at most the full cell length, hence at most the mesh, while its portion inside U_j is bounded by $e^{B_j}\mu_k(U_j)$; the omitted Lebesgue length is bounded in the same way by the mesh and $|U_j|$. Our choice of U_j, followed by large k, therefore gives (after harmlessly rescaling the sequence ε_j)

$$\sup_z \left| \int_0^z r_{k,j}\,\mathrm{d}\mu_k - z \right| \le 3\varepsilon_j.$$

Choose increasing integers K_j so that these conclusions hold for $k \ge K_j$ and additionally

$$B_j/K_j \le \varepsilon_j, \qquad C_j/K_j \le \varepsilon_j.$$

For $K_j \le k < K_{j+1}$ set $r_k = r_{k,j}$. Then $j = j(k) \to \infty$, so the cumulative error tends to zero and both estimates in (10.2) hold. The endpoint condition was built into every interpolation. □

10.2 The smooth exact gauge

Apply Theorem 10.1 to the measures in (9.18). Define a multiplier R on every sufficiently large shell by

$$R(2^k(1+z)) := r_k(z), \qquad 0 \le z \le 1. \tag{10.3}$$

Because $r_k = 1$ in neighborhoods of both endpoints, adjacent pieces agree with all derivatives near every power of two; hence R is globally smooth on a half-line. Put

$$h_0(x) := \frac{1}{g_1(x)}, \qquad h(x) := h_0(x)R(x), \qquad g_\infty(x) := \frac{1}{h(x)}. \tag{10.4}$$

In shell coordinates,

$$\frac{\partial}{\partial z}\log h_0(2^k(1+z)) = \frac{k+\kappa_1+a^{-1}\log(1+z)}{1+z} \ge \frac{k}{3} \tag{10.5}$$

for all large k, uniformly in z. Equation (10.2) therefore makes h eventually strictly increasing and g_∞ eventually strictly decreasing. It also gives

$$\log g_\infty(x) = \log g_1(x) + o(\log x). \tag{10.6}$$

Proposition 10.2 (Smooth exact arithmetic gauge)**.** *For every fixed sufficiently large X_0,*

$$\frac{1}{t-X_0}\int_{X_0}^t \frac{|\Phi(x)|}{g_\infty(x)}\,\mathrm{d}x \longrightarrow 1. \tag{10.7}$$

Proof. Let

$$C_k(z) := \int_0^z D_k(u) r_k(u)\,\mathrm{d}u.$$

By Theorem 10.1, $\sup_z |C_k(z) - z| \to 0$. If $t = 2^K(1+z)$, complete and partial shells give, up to one fixed initial contribution,

$$\int_{X_0}^t \frac{|\Phi(x)|}{g_\infty(x)}\,\mathrm{d}x = \sum_{k<K} 2^k C_k(1) + 2^K C_K(z).$$

Replacing $C_k(1)$ by 1 and $C_K(z)$ by z costs $o(2^K)$: indeed $\sum_{k<K} 2^k\epsilon_k = o(2^K)$ whenever $\epsilon_k \to 0$. The main term is $\sum_{k<K} 2^k + 2^K z = t + \mathcal{O}(1)$, after absorbing the fixed choice of initial shell. Division by $t - X_0$ proves (10.7). □

10.3 Real-analytic upgrade by Carleman–Hoischen approximation

We use the following precise classical input: if $F \in C^1(\mathbb{R})$ and $\epsilon : \mathbb{R} \to (0, \infty)$ is continuous, there is an entire function G, real on the real axis, such that

$$|G(x) - F(x)| < \epsilon(x), \qquad |G'(x) - F'(x)| < \epsilon(x) \quad (x \in \mathbb{R}). \tag{10.8}$$

This is Hoischen's derivative form of Carleman approximation; we use the formulation of Gauthier and Kienzle [8].

Extend h in (10.4) to a positive, strictly increasing C^1 function on $\mathbb{R}$. Choose a continuous $\delta(x) > 0$ with $\delta(x) \to 0$ as $x \to +\infty$ and apply (10.8) with

$$\epsilon(x) < \frac{1}{2} \min\{h(x), h'(x), \delta(x)h(x)\}. \tag{10.9}$$

The real entire approximant $\widetilde{h}$ is positive and strictly increasing on $\mathbb{R}$, and $\widetilde{h}/h \to 1$ at $+\infty$. Define

$$g_{\mathrm{ad}}(x) := \frac{1}{\widetilde{h}(x)}, \qquad x > 0. \tag{10.10}$$

Theorem 10.3 (Literal analytic solution of the original Cesaro problem). *The gauge g_{ad} is positive, real-analytic, and strictly decreasing. For every fixed $t_0 > 0$ it satisfies*

$$\frac{1}{t - t_0} \int_{t_0}^{t} \frac{|\Phi(x)|}{g_{\mathrm{ad}}(x)} \,\mathrm{d}x \longrightarrow 1, \tag{10.11}$$

and

$$\log g_{\mathrm{ad}}(x) = -\frac{(\log x)^2}{2\log 2} - \kappa_1 \log x + o(\log x). \tag{10.12}$$

Proof. The regularity and monotonicity follow from the construction. The rough scale follows from $\widetilde{h}/h \to 1$, (10.6), and the definition of g_1. For the integral, fix $\varepsilon > 0$ and choose X so that $|\widetilde{h}/h - 1| < \varepsilon$ for $x \geq X$. Since $1/g_{\mathrm{ad}} = \widetilde{h}$ and $1/g_\infty = h$,

$$\left| \int_X^t |\Phi(x)| \,(\widetilde{h}(x) - h(x)) \,\mathrm{d}x \right| \leq \varepsilon \int_X^t |\Phi(x)| \, h(x) \,\mathrm{d}x = \varepsilon(t + o(t))$$

by Theorem 10.2. The initial interval contributes $\mathcal{O}(1)$. Divide by t and then let $\varepsilon \downarrow 0$. □

Remark 10.4 (Why there is no contradiction). The natural profile is driven by the distribution function of an atomless singular measure, but the profile itself also contains the ordinary factor $1/y$ and is not itself a singular distribution function. The multiplier r_k corrects only finitely many cells at level k, while the number of cells tends to infinity. Its logarithm and logarithmic slope are $o(k)$, so the gauge remains decreasing, yet its normalized shell shape never converges and its distortion becomes unbounded. Thus neither Theorem 9.8 nor Theorem 9.9 applies.

Remark 10.5 (Higher derivatives). The natural gauge g_1 has alternating derivative signs eventually for each fixed order. The adaptive construction above guarantees positivity and strict decrease, but not one common tail on which every higher derivative has a prescribed sign. Preserving any chosen finite number of signs should be approachable by a higher-order Carleman–Hoischen theorem and a slower diagonal selection, but the required derivative error budget is not proved here. We record that strengthening as a frontier rather than as a theorem.

Lean currently formalizes the shell-linearity reduction, finite-partition measure equalization, and diagonal-selection core in `ShellLinearity`, `MeasureEqualization`, and `DiagonalSelection`. The RPF convergence, smooth gluing, and Carleman–Hoischen approximation remain explicitly identified analytic inputs.

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11 The multiplicatively natural logarithmic mean

The endpoint obstruction disappears if additive Cesaro averaging is replaced by logarithmic averaging. Each dyadic shell then has equal weight, and the final incomplete shell becomes negligible.

Theorem 11.1 (Unrestricted logarithmic mean). *There is a constant $A_1^{\log} > 0$ such that, for every fixed $t_0 > 1$,*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^{t} \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x} \longrightarrow A_1^{\log} \qquad (t \to \infty) \tag{11.1}$$

without any restriction on the endpoint. In the normalization of Theorem 5.1, if $f_1(x) = W_{\kappa_1}((x+1)/2)$, then

$$B_1 := \left(\int_0^1 h_1(x)\,dx \right) \nu_1\left(\frac{f_1}{1+\cdot} \right), \qquad A_1^{\log} := \frac{B_1}{\log 2}. \tag{11.2}$$

Numerically,

$$A_1^{\log} = 0.09386063575678126354\ldots. \tag{11.3}$$

Proof. Let

$$\beta_n = \int_{2^{n-1}}^{2^n} \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x}.$$

The master factorization and $x = 2^n y$, followed by $z = 2y - 1$, give

$$\beta_n = \rho_1^{-n} \int_0^1 \frac{f_1(z)}{1+z} \prod_{j=0}^{n-1} |\sin(\pi T^j z)|\,dz.$$

The spectral limit in Theorem 5.1 yields $\beta_n \to \beta_\infty$, where $\beta_\infty = (\int h_1)\nu_1(f_1/(1+\cdot))$. If $2^N \le t < 2^{N+1}$, the logarithmic integral equals $\sum_{n<N} \beta_n + O(1)$, while $\log(t/t_0) = N \log 2 + O(1)$. Ordinary Cesaro convergence of the sequence (β_n) proves (11.1) and (11.2). $\square$

12 The sharp uniform envelope

The L^∞ numerator rate can be proved by a sharp elementary inequality. Put

$$\lambda := \frac{\sqrt{3}}{2}. \tag{12.1}$$

Lemma 12.1 (Sharp two-step logarithmic inequality). *For every $x \in \mathbb{R}$,*

$$\frac{2}{3} \log s(x) + \frac{1}{3} \log s(Tx) \le \log \lambda, \tag{12.2}$$

with the usual interpretation at zeros. Equality holds precisely when $x \equiv 1/3$ or $2/3 \pmod 1$.

Proof. Cubing the desired inequality gives

$$s(x)^2 s(Tx) \le \lambda^3.$$

With $\theta = \pi x$ and $u = \sin^2\theta$,

$$s(x)^2 s(Tx) = 2\left|\sin^3\theta \cos\theta\right|,$$

whose square is $4u^3(1-u)$. The maximum of $u^3(1-u)$ on $[0,1]$ is $27/256$, attained at $u = 3/4$. Therefore the maximum of the original expression is $3\sqrt{3}/8 = \lambda^3$, with equality at $x \equiv 1/3, 2/3$. $\square$

Proposition 12.2 (Uniform lacunary-product bound). *For every $n \geq 1$,*

$$\lambda^n \leq \sup_{x\in[0,1]} \prod_{j=0}^{n-1} s(T^j x) \leq \lambda^{n-1}. \tag{12.3}$$

Consequently

$$\lim_{n\to\infty} \left\| \prod_{j=0}^{n-1} s \circ T^j \right\|_\infty^{1/n} = \lambda = \frac{\sqrt{3}}{2}. \tag{12.4}$$

Proof. At $x = 1/3$ the orbit alternates between $1/3$ and $2/3$, and every factor equals λ, proving the lower bound. For the upper bound, sum (12.2) for $x, Tx, \ldots, T^{n-2}x$. If $\phi = \log s$, the result is

$$3\sum_{j=0}^{n-2} \phi(T^j x) - \phi(x) + \phi(T^{n-1}x) \leq 3(n-1)\log\lambda.$$

After adding the final factor,

$$\sum_{j=0}^{n-1} \phi(T^j x) \leq (n-1)\log\lambda + \frac{1}{3}\phi(x) + \frac{2}{3}\phi(T^{n-1}x) \leq (n-1)\log\lambda,$$

because $\phi \leq 0$. Exponentiation gives the upper bound. □

This is the classical Gelfond exponent for the Thue–Morse polynomial: by (3.14), the unnormalized polynomial grows like $3^{n/2}$ in supremum norm. The ergodic-optimization formulation and its generalizations are developed in [6].

Theorem 12.3 (Sharp global upper envelope). *Define*

$$\boxed{\kappa_\infty := \frac{\log\pi + a/2 - \log(\sqrt{3}/2)}{a} = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2} = 2.35901487911174070 73\ldots.} \tag{12.5}$$

Then there is a constant $C < \infty$ such that

$$|\Phi(x)| \leq CE_{\kappa_\infty}(x), \qquad x \geq 1. \tag{12.6}$$

This exponent is sharp. In fact, for every $n \geq 0$,

$$|\Phi(2^n \cdot 2/3)| = C_* E_{\kappa_\infty}(2^n \cdot 2/3), \tag{12.7}$$

where

$$C_* = W_{\kappa_\infty}(2/3) > 0. \tag{12.8}$$

Numerically, $C_* = 0.139129774734829\ldots$; the exact content is the preceding value of W_{κ_∞}, not a certified decimal enclosure.

Proof. In (3.12), the definition of κ_∞ makes

$$\exp(n(\kappa_\infty a - \log\pi - a/2)) = \lambda^{-n}.$$

After the substitution $x = Ty$, the product is of the form bounded in (12.3); therefore

$$\frac{|\Phi(2^n y)|}{E_{\kappa_\infty}(2^n y)} \leq \lambda^{-1} \max_{1/2\leq y\leq 1} W_{\kappa_\infty}(y).$$

This proves the uniform upper bound. At $y = 2/3$, the orbit alternates between $2/3$ and $1/3$, so $B_n(2/3) = \lambda^n$ exactly. Equation (3.12) then reduces to (12.7). □

Corollary 12.4 (Universal logarithmic order). *With the convention* $\log 0 = -\infty$*,*

$$\limsup_{x\to\infty} \frac{\log|\Phi(x)|}{(\log x)^2} = -\frac{1}{2\log 2}, \qquad \liminf_{x\to\infty} \frac{\log|\Phi(x)|}{(\log x)^2} = -\infty. \tag{12.9}$$

Equivalently, the coefficient $1/(2\log 2)$ *is the sharp global log-square order, but no positive two-sided pointwise equivalent exists.*

Proof. The upper bound in the limsup follows from (12.6); the exact period-two ray (12.7) gives the reverse inequality. The liminf is $-\infty$ because Φ vanishes at every positive integer (and also because one may approach those zeros arbitrarily closely). □

Corollary 12.5 (Sharp smoothness class). *For every* $M > 0$*,* $\Phi(x) = \mathcal{O}(x^{-M})$ *as* $x \to \infty$*. On the other hand, for every* $c > 0$*,* $\mathrm{e}^{cx}\,|\Phi(x)|$ *is unbounded. Thus the transform decays faster than every power but not at any fixed exponential rate.*

Proof. The first statement follows from (12.6), because the negative quadratic in $\log x$ dominates $-M\log x$. The exact peak ray (12.7) decays only like $\exp(-\mathcal{O}((\log x)^2))$, which is much larger than e^{-cx} along that subsequence. □

13 The raw dyadic-annulus maximum and its Lambert boundary layer

The sharp pointwise envelope of Theorem 12.3 is evaluated at the actual argument x. A different question is to compare every value in the shell $[2^k, 2^{k+1}]$ with the envelope at its *left endpoint*. The maximizer then moves toward that endpoint and pays a second, slower log-normal cost. This is the main result of Wave 1, Document 2 [22]; we reproduce the proof because it is the most delicate asymptotic in the corpus.

Put

$$\mathfrak{M}_k := \max_{2^k \le x \le 2^{k+1}} |\Phi(x)|, \qquad n := k+1, \qquad \beta := \log\frac{\sqrt{3}}{2} < 0. \tag{13.1}$$

Also define the tail factor

$$\mathcal{H}(y) := \prod_{m=1}^{\infty} \left|\operatorname{sinc}\left(\frac{\pi y}{2^m}\right)\right|, \qquad y \in \mathbb{R}.$$

Thus $\mathcal{H}(y) = |\Phi(y/2)|$ globally. In particular it is continuous and bounded; on compact sets avoiding the nonzero even integers it is bounded away from zero.

13.1 Exact reduction and transition coordinates

Writing $x = 2^k(1+\theta)$, $0 \le \theta \le 1$, gives

$$|\Phi(2^k(1+\theta))| = \frac{2^{-k(k+1)/2}}{\pi^n}\mathcal{H}(1+\theta)(1+\theta)^{-n}Q_n(\theta), \tag{13.2}$$

where

$$Q_n(\theta) = \prod_{j=0}^{n-1} |\sin(\pi 2^j\theta)|.$$

Hence

$$\mathfrak{M}_k = \pi^{-n}2^{-k(k+1)/2}A_n, \quad A_n := \sup_{0\le\theta\le 1} \mathcal{H}(1+\theta)(1+\theta)^{-n}Q_n(\theta). \tag{13.3}$$

The weight $(1+\theta)^{-n}$ pushes the maximum toward 0, where Q_n vanishes. For $0<\theta\le 1/2$, write uniquely up to endpoints

$$\theta = 2^{-r}s, \qquad r\in\mathbb{Z}_{\ge 0}, \quad s\in I := [1/4,1/2]. \tag{13.4}$$

If $r<n$, then

$$Q_n(2^{-r}s) = R_r(s)Q_{n-r}(s), \quad R_r(s) = \prod_{h=1}^{r} \sin\Big(\frac{\pi s}{2^h}\Big). \tag{13.5}$$

Uniformly for $s\in I$,

$$\log R_r(s) = -\frac{a}{2}r^2 + r\left(\log(\pi s) - \frac{a}{2}\right) + O(1). \tag{13.6}$$

The remaining tail satisfies $Q_m(s)\le C\mathrm{e}^{m\beta}$ by Theorem 12.2.

For a matching lower bound one needs near-maximizing tails at essentially arbitrary s.

Lemma 13.1 (Dense preimages of the maximizing cycle). *There is $C>0$ such that for every $s_0\in I$ and $q\ge 1$ one can choose $s_q\in I$ with*

$$|s_q - s_0| \le 2^{-q}, \qquad T^q s_q \in \{1/3, 2/3\}, \tag{13.7}$$

and, for every $m\ge q$,

$$Q_m(s_q) \ge \exp(m\beta - Cq^2). \tag{13.8}$$

Proof. The q-fold preimages $(j+1/3)/2^q$ have spacing 2^{-q}; together with the preimages of $2/3$ they approximate every point of I as in (13.7). Before time q, every orbit point is a nonintegral rational with reduced denominator at most $3\cdot 2^{q-j}$. Therefore

$$|\sin(\pi T^j s_q)| \ge 2\,\mathrm{dist}(T^j s_q,\mathbb{Z}) \ge \frac{2}{3\cdot 2^{q-j}}.$$

The first q factors lose at most $\exp(O(q^2))$. Thereafter the orbit alternates between $1/3$ and $2/3$, and every factor equals $\sqrt{3}/2 = \mathrm{e}^{\beta}$. □

Define the finite-dimensional objective

$$\mathcal{F}_n(r,s) = -\frac{a}{2}r^2 + r\left(\log(\pi s) - \frac{a}{2} - \beta\right) - n\log(1+s2^{-r}). \tag{13.9}$$

Lemma 13.2 (Reduction to the boundary-layer objective). *As $n\to\infty$,*

$$\log A_n = n\beta + \sup_{\substack{r\in\mathbb{Z}_{\ge 0}\\ s\in I}} \mathcal{F}_n(r,s) + O((\log\log n)^2). \tag{13.10}$$

Proof. First take $0<\theta\le 1/2$, write $\theta=2^{-r}s$, and suppose $r<n$. Equations (13.5)–(13.6) and the uniform Gelfond bound give, uniformly in r, s,

$$\log\big(\mathcal{H}(1+\theta)(1+\theta)^{-n}Q_n(\theta)\big) \le n\beta + \mathcal{F}_n(r,s) + C. \tag{13.11}$$

Here $\mathcal{H}$ is bounded on $[1,2]$. If $\theta\ge 1/2$, the left side is at most $-n\log(3/2)+O(1)$. If $r\ge n$, the elementary bounds $|\sin u|\le |u|$ give

$$\log Q_n(2^{-r}s) \le n\log(\pi s) + a\frac{n(n-1)}{2} - anr,$$

so this range is $-\Omega(n^2)$. Both are exponentially inferior to the admissible trial $r=\lfloor \log_2 n\rfloor$, $s=1/3$, whose tail lies exactly on the maximizing two-cycle and whose logarithm is $n\beta - O((\log n)^2)$. This proves that every phase within $O((\log\log n)^2)$ of the maximum has $r<n$.

The same comparison yields the two uniform localization estimates used below. Since $s \in [1/4, 1/2]$,

$$\mathcal{F}_n(r,s) \le -\frac{a}{2}r^2 + C_1 r, \qquad \log(1+s2^{-r}) \ge \frac{1}{2}s2^{-r}.$$

Comparison with the trial value first gives $r = O(\log n)$ and then, after retaining the last negative term in $\mathcal{F}_n$,

$$ns2^{-r} = O((\log n)^2). \tag{13.12}$$

Thus (13.11) gives the desired upper bound by taking the supremum of $\mathcal{F}_n$.

For the reverse inequality choose a localized pair (r,s) within one of the supremum of $\mathcal{F}_n$ and put $q = \lceil 4\log_2 \log(n + \mathrm{e}^{\mathrm{e}})\rceil$. For large n we have $n - r \ge q$. Choose s_q by Theorem 13.1. In the localized region

$$\left|\frac{\partial \mathcal{F}_n}{\partial s}\right| \le 4r + n2^{-r} = O((\log n)^2),$$

so $|\mathcal{F}_n(r,s_q) - \mathcal{F}_n(r,s)| = O(1)$. The dense-preimage lower bound gives

$$\log Q_{n-r}(s_q) \ge (n-r)\beta - Cq^2,$$

while (13.6) supplies the matching prefix and $\mathcal{H}(1 + 2^{-r}s_q)$ is bounded below because its argument tends to 1. Consequently the resulting phase has logarithm at least

$$n\beta + \sup_{r,s} \mathcal{F}_n(r,s) - O(q^2).$$

Since $q^2 = O((\log\log n)^2)$, this completes both sides of (13.10). □

13.2 Lambert optimization

Set

$$C_0 := \log \pi - \frac{a}{2} - \beta = \log\left(\pi\sqrt{\frac{2}{3}}\right), \qquad c := a\mathrm{e}^{-C_0} = \frac{a}{\pi}\sqrt{\frac{3}{2}}. \tag{13.13}$$

At the localized maximum, (13.12) implies

$$n\left|\log(1+s2^{-r}) - s2^{-r}\right| = O\left(\frac{(\log n)^4}{n}\right) = o(1).$$

Put $A = C_0 + \log s$ and $z = ar - A$. Then

$$-\frac{a}{2}r^2 + rA - ns2^{-r} = -\frac{z^2}{2a} - n\mathrm{e}^{-C_0}\mathrm{e}^{-z} + \frac{A^2}{2a}. \tag{13.14}$$

As s traverses $[1/4, 1/2]$, the interval of possible A has length exactly a. Successive integer values of $r \ge 0$ therefore make the corresponding z-intervals tile a half-line. That half-line contains the maximizer below for all large n. Since $A^2/(2a) = O(1)$, the supremum is, up to $O(1)$, the real maximum of

$$\phi_n(z) = -\frac{z^2}{2a} - n\mathrm{e}^{-C_0}\mathrm{e}^{-z}.$$

Its critical equation is $z\mathrm{e}^z = cn$, so the maximizer is $w_n = W(cn)$ and we use the principal real branch of W throughout [9]. Thus

$$\max \phi_n = -\frac{w_n^2 + 2w_n}{2a}. \tag{13.15}$$

Theorem 13.3 (Sharp dyadic-annulus maximum). *Let $w_k = W(c(k+1))$, with c from (13.13). Then*

$$\boxed{\log \mathfrak{M}_k = -\frac{a}{2}k(k+1) - (k+1)\log\pi + (k+1)\beta - \frac{w_k^2 + 2w_k}{2a} + O((\log\log k)^2).} \tag{13.16}$$

Equivalently,

$$\log \mathfrak{M}_k = -\frac{a}{2}k^2 - a\kappa_\infty k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2a} + O((\log\log k)^2). \tag{13.17}$$

Any maximizing phase θ_k satisfies

$$\theta_k = \frac{W(c(k+1))}{a(k+1)} \mathrm{e}^{o(1)}. \tag{13.18}$$

Proof. Combine (13.3), Theorem 13.2, and (13.15). The compact form uses $a\kappa_\infty = a/2 + \log\pi - \beta$.

For the location statement, (13.11) and the matching lower bound show that the z associated with any maximizing phase satisfies $\phi_n(w_n) - \phi_n(z) = O((\log\log n)^2)$. Writing $z = w_n + u$ and using $n\mathrm{e}^{-C_0 - w_n} = w_n/a$, one has

$$\phi_n'(w_n + u) = \frac{w_n(\mathrm{e}^{-u} - 1) - u}{a}.$$

For $u \le 0$ this gives a loss at least $w_n u^2/(2a)$; for $0 \le u \le 1$ it gives the same bound up to an absolute constant. A displacement $u \ge 1$ already costs a fixed positive multiple of w_n. Since $w_n \asymp \log n$ while $(\log\log n)^2 = o(w_n)$, every maximizer therefore satisfies

$$z = w_n + O\left(\frac{\log\log n}{\sqrt{w_n}}\right) = w_n + o(1).$$

Since $\theta = s2^{-r} = \mathrm{e}^{-C_0 - z}$ and $n\mathrm{e}^{-C_0 - w_n} = w_n/a$, this gives (13.18). □

Let $R = 2^k$, $L = \log R$, and $d = \pi^{-1}\sqrt{3/2}$. Since $c(k+1) = d(L+a)$, (13.17) becomes

$$\log \max_{R \le x \le 2R} |\Phi(x)| = -\frac{L^2}{2a} - \kappa_\infty L - \frac{W(d(L+a))^2 + 2W(d(L+a))}{2a} + O((\log\log\log R)^2). \tag{13.19}$$

The extra term begins

$$-\frac{(\log\log R)^2}{2a} + \frac{\log\log R \, \log\log\log R}{a} + O(\log\log R). \tag{13.20}$$

Thus the annular maximum is smaller than the left-endpoint envelope $E_{\kappa_\infty}(R)$ by a second log-normal factor, even though the pointwise-envelope rate is realized exactly up to C_* on the fixed ray $x = 2^n(2/3)$. There is no contradiction: the fixed ray is a constant fraction into its shell, whereas the raw annular maximizer migrates toward the left boundary according to (13.18).

14 Exceptional extrema near dyadic boundaries

The sharp envelope in Theorem 12.3 describes the largest possible values in a shell, but it does not make every local extremum comparable to one constant. The deepest and most explicit obstruction occurs immediately after powers of two.

14.1 The first lobe after a fixed odd multiple

Theorem 14.1 (Peaks after an odd multiple of a power of two)**.** *Fix an odd positive integer m. Write the unique maximizer in $(m2^k, m2^k+1)$ as $m2^k+t_k$, $0<t_k<1$. Then*

$$(k+1)(1-t_k) \longrightarrow 1, \tag{14.1}$$

and

$$\boxed{M_{m2^k} \sim \frac{\mathcal{H}(1)\,|\mathcal{H}(m)|}{\mathrm{e}(k+1)}(m2^k)^{-(k+1)}.} \tag{14.2}$$

In particular,

$$M_{2^k} \sim \frac{\mathcal{H}(1)^2}{\mathrm{e}(k+1)}2^{-k(k+1)}. \tag{14.3}$$

Proof. Put $N=m2^k$. Splitting the product at $h=k$ and using the integral-periodicity of the sine numerators gives the exact zero-safe identity

$$|\Phi(N+t)| = \left(\frac{t}{N+t}\right)^{k+1} \frac{|\Phi(t)|}{|\Phi(t/2^{k+1})|}\mathcal{H}\left(m+\frac{t}{2^k}\right), \qquad 0\le t\le 1. \tag{14.4}$$

Indeed, the first $k+1$ sine numerators have the same absolute values as those of $\Phi(t)$, the denominator quotient is $(t/(N+t))^{k+1}$, and reindexing the two tails produces the remaining quotient and $\mathcal{H}$ factor.

Uniformly for $0\le t\le 1$, $\Phi(t/2^{k+1})\to 1$, $\mathcal{H}(m+t/2^k)\to\mathcal{H}(m)$, and $(1+t/N)^{-(k+1)}\to 1$. Hence

$$|\Phi(N+t)| = N^{-(k+1)}\,|\mathcal{H}(m)|\,t^{k+1}\Phi(t)(1+o(1)) \tag{14.5}$$

uniformly on $[0,1]$. On any fixed neighborhood of 1 the logarithmic derivative of the omitted multiplier is $\mathcal{O}((k+1)/N)+\mathcal{O}(2^{-k})=o(1)$, by local uniform convergence of the product and its derivative. Thus the critical-point equation below is perturbed only by $o(1)$. On $[0,1)$ the function $\Phi(t)$ is positive, and Theorem 3.3 at the simple zero 1 gives

$$\Phi(t) = \mathcal{H}(1)(1-t)(1+\mathcal{O}(1-t)). \tag{14.6}$$

Let $p=k+1$. A maximizer of $t^p\Phi(t)$ must tend to 1: values on $[0,1-\delta]$ are exponentially smaller than the value at $1-1/p$. At its interior maximizer t_p, logarithmic differentiation gives $p/t_p+\Phi'(t_p)/\Phi(t_p)=0$. Equation (14.6) implies $(1-t)\Phi'(t)/\Phi(t)\to -1$, so $p(1-t_p)\to 1$. Consequently $t_p^p\to\mathrm{e}^{-1}$ and $p\Phi(t_p)\to\mathcal{H}(1)$. For the exact lobe expression, logarithmic differentiation adds only the $o(1)$ term estimated above, so its critical equation gives $p(1-t_k)=1+o(1)$ by the same argument. Substitution into the uniform value asymptotic (14.5) proves (14.1)–(14.2). Taking $m=1$ gives (14.3). □

14.2 Every fixed offset after a power of two

Theorem 14.2 (Dyadic valleys: exact extremal asymptotic)**.** *Fix an integer $r\ge 0$, and put*

$$b := r+1, \qquad d := 1+\nu_2(b), \qquad N := 2^{n-1}.$$

Let ξ_{N+r} be the unique critical point in $(N+r, N+r+1)$ and $M_{N+r}=|\Phi(\xi_{N+r})|$. Then, as $n\to\infty$,

$$\xi_{N+r} = N+b-\frac{db}{n}+o(n^{-1}), \tag{14.7}$$

$$M_{N+r} \sim \left|\Phi(1/2)\Phi(b/2^d)\right| \left(\frac{b}{N}\right)^n \left(\frac{d}{\mathrm{e}n}\right)^d. \tag{14.8}$$

In particular,

$$\log M_{N+r} = -\frac{(\log N)^2}{\log 2} + \mathcal{O}(\log N). \tag{14.9}$$

The leading quadratic coefficient in (14.9) *is twice the coefficient in* E_κ.

Proof. Write $x = N + z$, $r < z < b$, in the exact boundary identity (3.16). Uniformly for $z \in [r, b]$,

$$\frac{\left|\Phi(\frac{1}{2} + z/(2N))\right|}{|\Phi(z/(2N))|} \longrightarrow |\Phi(1/2)|\,, \qquad \left(1 + \frac{z}{N}\right)^{-n} \longrightarrow 1, \tag{14.10}$$

because $n/N \to 0$. Hence

$$M_{N+r} = |\Phi(1/2)|\, N^{-n} \max_{r \le z \le b} \left(z^n\, |\Phi(z)|\right)(1 + o(1)). \tag{14.11}$$

By (3.8), as $z \uparrow b$,

$$|\Phi(z)| = D_b(b-z)^d(1 + \mathcal{O}(b-z)), \qquad D_b = b^{-d}\left|\Phi(b/2^d)\right|.$$

Choose $\delta_0 > 0$ so small that the local expansion bounds $|\Phi(z)|$ above and below by fixed positive multiples of $(b-z)^d$ on $[b-\delta_0, b]$. The trial point $z = b - db/n$ has size at least $c\, b^{n+d} n^{-d}$ for large n, whereas $z \le b - \delta_0$ is exponentially smaller. On the remaining interval put $u = n(b-z)/b$. Since $(1 - u/n)^n \le \mathrm{e}^{-u}$,

$$z^n\, |\Phi(z)| \le C b^{n+d} n^{-d} u^d \mathrm{e}^{-u}.$$

After $U > d$ is chosen large, the supremum of $u^d e^{-u}$ over $u \ge U$ is smaller than the trial-point lower constant. Thus every maximizer has u in one fixed compact interval. Uniformly there,

$$z^n\, |\Phi(z)| = D_b b^{n+d} n^{-d}\, u^d \left(1 - \frac{u}{n}\right)^n (1 + o(1)) \longrightarrow D_b b^{n+d} n^{-d} u^d \mathrm{e}^{-u}.$$

The function $u^d \mathrm{e}^{-u}$ has its unique maximum at $u = d$, with value $d^d \mathrm{e}^{-d}$. Uniform convergence on the compact interval just obtained therefore gives

$$b - z_{n,r} \sim \frac{db}{n}$$

and

$$\max z^n\, |\Phi(z)| \sim D_b b^{n+d} n^{-d} d^d \mathrm{e}^{-d}.$$

Insert $D_b = b^{-d}\left|\Phi(b/2^d)\right|$ into (14.11) to obtain (14.8); the location statement is (14.7). Finally, $-n \log N = -(\log N)^2/a + \mathcal{O}(\log N)$, proving (14.9). $\square$

For example, in the first interval after $N = 2^{n-1}$, $r = 0$, $b = d = 1$, and

$$M_N \sim \frac{\Phi(1/2)^2}{\mathrm{e}n} N^{-n}, \qquad \xi_N = N + 1 - \frac{1}{n} + o(n^{-1}). \tag{14.12}$$

Numerically, $\Phi(1/2) = 0.553771275888810\ldots$.

14.3 Consequences for the sequence of extrema

Corollary 14.3 (The upper envelope is sharp but not two-sided). *For* $M_m = \max_{[m,m+1]} |\Phi|$,

$$0 < \limsup_{m\to\infty} \frac{M_m}{E_{\kappa_\infty}(m)} < \infty, \qquad \liminf_{m\to\infty} \frac{M_m}{E_{\kappa_\infty}(m)} = 0. \tag{14.13}$$

Therefore no constant multiple of the sharp uniform envelope makes all local extrema approach one nonzero amplitude.

Proof. The upper bound follows from (12.6). Let $x_n = 2^n(2/3)$ and $m_n = \lfloor x_n \rfloor$. The local maximum in the containing unit interval is at least $|\Phi(x_n)|$, and (12.7) gives a positive lower bound after division by $E_{\kappa_\infty}(m_n)$; changing the argument by less than one changes E_{κ_∞} by a factor tending to one. This proves the positive limsup.

For the liminf, take $m = N + r$ with fixed r. Comparing (14.9) with $\log E_{\kappa_\infty}(N) = -(\log N)^2/(2a) + \mathcal{O}(\log N)$ shows that the quotient tends to zero. $\square$

This is a stronger obstruction than the integer zeros alone: even after replacing the function by the absolute values of its unique local extrema, a constant-amplitude normalization does not exist.

15 A continuum of intermediate boundary rates

The boundary identity also explains why many intermediate effective decay rates appear. Suppose $N = 2^{n-1}$, $z = z_n = o(N)$, and $z_n \to \infty$. From (3.16), provided the tail ratio remains bounded away from zero and infinity,

$$\log|\Phi(N + z_n)| = \log|\Phi(z_n)| + n\log\frac{z_n}{N} + o(n). \tag{15.1}$$

If $\log z_n \sim \alpha \log N$ with $0 \le \alpha < 1$ and z_n itself follows a typical fixed-phase decay law, then substitution of (6.4) into (15.1) gives the quadratic coefficient

$$\log|\Phi(N + z_n)| = -\frac{1 - \alpha + \alpha^2/2}{a}(\log N)^2 + \mathcal{O}(\log N) + o((\log N)^2). \tag{15.2}$$

The coefficient decreases continuously from $1/a$ at $\alpha = 0$ to $1/(2a)$ as $\alpha \uparrow 1$. This calculation is conditional only on the specified decay law for z_n and makes precise the recursive origin of the broad scatter visible in plots of the extrema.

16 Integer geometric ratios and a general variance law

The dyadic product belongs to a natural one-parameter family. Let $b > 1$ and define

$$\Phi_b(z) := \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^n}\right). \tag{16.1}$$

Local uniform convergence follows from $\operatorname{sinc} w = 1 - w^2/6 + \mathcal{O}(w^4)$ and $\sum b^{-2n} < \infty$. The product is therefore entire for every real $b > 1$.

16.1 Global and arbitrary-radius Pochhammer factorizations

Proposition 16.1 (Global spectral Pochhammer factorization). *For every $b > 1$ and $z \in \mathbb{C}$,*

$$\boxed{\Phi_b(z) = \prod_{\ell=1}^{\infty}\left(\frac{z^2}{\ell^2}; b^{-2}\right)_\infty}. \tag{16.2}$$

More generally, for the tail $R_{b,m}(z) := \prod_{n=m}^{\infty} \operatorname{sinc}(\pi z/b^n)$,

$$R_{b,m}(z) = \prod_{\ell=1}^{\infty}\left(\frac{z^2}{\ell^2 b^{2m}}; b^{-2}\right)_\infty. \tag{16.3}$$

Both products converge locally uniformly.

Proof. Euler's sine product gives

$$\operatorname{sinc}\left(\frac{\pi z}{b^n}\right) = \prod_{\ell=1}^{\infty}\left(1 - \frac{z^2}{\ell^2 b^{2n}}\right).$$

The perturbations from 1 are absolutely summable on every compact set because

$$\sum_{n=0}^{\infty}\sum_{\ell=1}^{\infty} \frac{|z|^2}{\ell^2 b^{2n}} = \frac{|z|^2\zeta(2)}{1-b^{-2}} < \infty.$$

Hence the double product may be rearranged, including when one factor is zero. Grouping by ℓ proves (16.2); replacing z by z/b^m proves (16.3). □

Corollary 16.2 (Zero divisor for an integer ratio)**.** *If $b \ge 2$ is an integer, the nonzero zeros of Φ_b are precisely the nonzero integers. The zero at $M \neq 0$ has multiplicity*

$$1 + \nu_b(|M|), \qquad \nu_b(M) := \max\{j \ge 0 : b^j \mid M\}.$$

Proof. In (16.2), a zero at M occurs once for every representation $|M| = \ell b^n$ with $\ell \ge 1$, $n \ge 0$. These representations correspond exactly to $0 \le n \le \nu_b(|M|)$. □

For an integer $J \ge 1$, put

$$H_J^{(s)} := \sum_{\ell=1}^{J} \ell^{-s}, \qquad d_{J,k} := \frac{\zeta(2k) - H_J^{(2k)}}{k}.$$

Theorem 16.3 (Arbitrary-radius finite-zero extraction)**.** *Let $b > 1$, let $m \ge 0$ be an integer, and let $J \ge 1$ be an integer. If $|z| < (J+1)b^m$, then*

$$\begin{aligned} R_{b,m}(z) = {} & \prod_{\ell=1}^{J}\left(\frac{z^2}{\ell^2 b^{2m}}; b^{-2}\right)_\infty \\ & \times \exp\left(-\sum_{k=1}^{\infty} \frac{d_{J,k}}{1-b^{-2k}}\left(\frac{z}{b^m}\right)^{2k}\right). \end{aligned} \tag{16.4}$$

The exponent series has radius exactly $(J+1)b^m$.

Proof. Separate the factors $1 \le \ell \le J$ in (16.3). With $u = z/b^m$, the remaining logarithmic series is absolutely convergent for $|u| < J+1$, and

$$\begin{aligned} & \prod_{\ell=J+1}^{\infty}\prod_{j=0}^{\infty}\left(1 - \frac{u^2}{\ell^2 b^{2j}}\right) \\ & = \exp\left[-\sum_{k=1}^{\infty}\frac{u^{2k}}{k}\left(\sum_{\ell=J+1}^{\infty}\ell^{-2k}\right)\left(\sum_{j=0}^{\infty} b^{-2jk}\right)\right], \end{aligned}$$

which is (16.4). Finally $\zeta(2k) - H_J^{(2k)} = (J+1)^{-2k}(1+o(1))$, so the root test gives the stated exact radius. The next unextracted zeros at $u = \pm(J+1)$ give the same geometric explanation. □

The dyadic theorem 3.1 is the case $b = 2$, $J = 1$. Thus the original idea admits two independent accuracy controls: increase the tail index m, extract more zero layers J, or balance both. This is a bounded-range evaluation mechanism, not a substitute for the shell dynamics as $z \to \infty$.

16.2 Exact integer-b shell and ray laws

Now assume $b \geq 2$ is an integer. Let $T_b(t) = bt \pmod 1$ on $[0,1)$ and $\phi(t) = \log|\sin(\pi t)|$, with $\phi(0) = -\infty$.

Theorem 16.4 (Exact b-adic shell identity). *Let $b \geq 2$ be an integer, let $N \geq 0$, $1 \leq y < b$, and $z = \{y\}$. In the extended real line,*

$$\begin{aligned} \log|\Phi_b(b^N y)| = {} & \log|\Phi_b(y)| - N\log(\pi y) - \frac{\log b}{2} N(N+1) \\ & + \sum_{k=1}^{N} \phi(T_b^k z). \end{aligned} \tag{16.5}$$

At a zero, both sides are interpreted as $-\infty$.

Proof. Reindexing (16.1) gives

$$\Phi_b(b^N y) = \Phi_b(y) \prod_{k=1}^{N} \operatorname{sinc}(\pi b^k y).$$

Because b is an integer, $|\sin(\pi b^k y)| = |\sin(\pi T_b^k z)|$. Take absolute values and logarithms and use $\sum_{k=1}^{N} k \log b = \frac{1}{2}N(N+1)\log b$. □

Theorem 16.5 (Fixed-ray and typical integer-ratio decay). *Let $b \geq 2$ be an integer and $1 \leq y < b$. Assume $\Phi_b(y) \neq 0$ and that*

$$\alpha_b(y) := \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \phi(T_b^k z)$$

exists and is finite. For $x_N = b^N y$,

$$\log|\Phi_b(x_N)| = -\frac{(\log x_N)^2}{2\log b} - \kappa_b(\alpha_b(y)) \log x_N + o(\log x_N), \tag{16.6}$$

where

$$\kappa_b(\alpha) := \frac{\log \pi + \frac{1}{2}\log b - \alpha}{\log b}. \tag{16.7}$$

For Lebesgue-almost every $y \in (1,b)$,

$$\boxed{\kappa_{\text{typ}}(b) = \frac{1}{2} + \log_b(2\pi).} \tag{16.8}$$

Proof. Put $B = \log b$. If the Birkhoff average is $\alpha_b(y)$, then (16.5) is

$$-\frac{B}{2}N^2 + N\left(\alpha_b(y) - \log(\pi y) - \frac{B}{2}\right) + o(N)$$

up to the constant $\log|\Phi_b(y)|$. Substitute $N = (\log x_N - \log y)/B$. The two mixed terms involving $\log y \, \log x_N$ cancel, leaving (16.6)–(16.7).

The map T_b preserves Lebesgue measure and is ergodic, while $\phi \in L^1(0,1)$ and $\int_0^1 \phi = -\log 2$. Birkhoff's theorem gives $\alpha_b(y) = -\log 2$ for almost every mantissa, and substitution yields (16.8). For $b = 2$ this is $\frac{3}{2} + \log_2 \pi = \kappa_0$. □

The centered form makes the mechanism especially transparent. With $\psi(t) = \log|2\sin(\pi t)|$ and $C_b(y) = \log|\Phi_b(y)| + (\log y)^2/(2\log b) + \kappa_{\mathrm{typ}}(b)\log y$, one has, away from zeros,

$$\log|\Phi_b(b^N y)| + \frac{(\log(b^N y))^2}{2\log b} + \kappa_{\mathrm{typ}}(b)\log(b^N y) = \sum_{k=1}^{N} \psi(T_b^k z) + C_b(y). \tag{16.9}$$

16.3 Exact covariance, variance, CLT, and LIL for every integer ratio

Theorem 16.6 (Exact b-adic logarithmic variance). *Let $b \geq 2$ be an integer and let $S_N(t) = \sum_{j=0}^{N-1} \psi(T_b^j t)$. Then, for $j \geq 0$,*

$$\int_0^1 \psi(t)\psi(T_b^j t)\,\mathrm{d}t = \frac{\pi^2}{12 b^j}, \tag{16.10}$$

and

$$\begin{aligned}\int_0^1 S_N(t)^2\,\mathrm{d}t &= \frac{\pi^2}{12}\left[N + 2\sum_{j=1}^{N-1}(N-j)b^{-j}\right] \\ &= \frac{\pi^2}{12}\left[\frac{b+1}{b-1}N - \frac{2b}{(b-1)^2} + \frac{2b^{1-N}}{(b-1)^2}\right].\end{aligned} \tag{16.11}$$

Thus the asymptotic variance is

$$\boxed{\sigma_b^2 = \frac{\pi^2}{12}\frac{b+1}{b-1}.} \tag{16.12}$$

Proof. In $L^2(0,1)$,

$$\psi(t) = -\sum_{n=1}^{\infty} \frac{\cos(2\pi n t)}{n}.$$

After replacing t by $b^j t$, cosine orthogonality leaves only pairs $n = mb^j$, so

$$\int_0^1 \psi(t)\psi(T_b^j t)\,\mathrm{d}t = \frac{1}{2}\sum_{m=1}^{\infty} \frac{1}{(mb^j)m} = \frac{\pi^2}{12 b^j}.$$

Stationarity makes the covariance depend only on the lag. Summing the diagonal and two off-diagonal halves gives the first line of (16.11). The finite geometric-arithmetic sum is

$$\sum_{j=1}^{N-1}(N-j)b^{-j} = \frac{N}{b-1} - \frac{b}{(b-1)^2} + \frac{b^{1-N}}{(b-1)^2},$$

which proves the second line and (16.12). □

There is also an exact martingale decomposition. For the Lebesgue transfer operator

$$(\mathcal{P}_b f)(x) = \frac{1}{b}\sum_{r=0}^{b-1} f\left(\frac{x+r}{b}\right),$$

the Fourier series gives $\mathcal{P}_b\psi = \psi/b$. Hence, with

$$h = \frac{\psi}{b-1}, \qquad d = \psi + h - h\circ T_b = \frac{b\psi - \psi\circ T_b}{b-1},$$

one has $\mathcal{P}_b d = 0$ and

$$S_N = \sum_{j=0}^{N-1} d \circ T_b^j - h + h \circ T_b^N, \qquad \|d\|_2^2 = \sigma_b^2. \tag{16.13}$$

Corollary 16.7 (Integer-ratio CLT and LIL; imported martingale theorem). *For every integer $b \geq 2$, put $S_N(t) = \sum_{j=0}^{N-1} \psi(T_b^j t)$ and $\sigma_b^2 = \frac{\pi^2}{12}\frac{b+1}{b-1}$. Then*

$$\frac{S_N}{\sigma_b\sqrt{N}} \Longrightarrow \mathcal{N}(0,1), \qquad \limsup_{N\to\infty} \frac{S_N}{\sqrt{2\sigma_b^2 N \log\log N}} = 1, \qquad \liminf_{N\to\infty} \frac{S_N}{\sqrt{2\sigma_b^2 N \log\log N}} = -1$$

for Lebesgue-almost every t. At $b = 2$, (16.11) is exactly Theorem 7.1, and $\sigma_2^2 = \pi^2/4$ recovers the corrected dyadic constant.

Proof. The sequence $(d \circ T_b^j)$ is a stationary ergodic reverse martingale difference. Its predictable quadratic variation divided by N converges almost surely to $\|d\|_2^2 = \sigma_b^2$ by the ergodic theorem. The logarithmic singularities give moments of every finite order and exponential tails, hence the Borel–Cantelli maximal bound and conditional Lindeberg summability checked in the proof of Theorem 7.2. The same imported stationary martingale CLT and Stout LIL therefore apply. Finally the coboundary $-h + h \circ T_b^N$ is negligible on both normalizations by the identical exponential-tail argument. $\square$

The global dyadic Pochhammer identity is represented in the current Lean source by `RvachevPochhammerFactorization`; the entire contracting-ratio product is developed in `GeometricReciprocalGamma`; and the case $b = 2$ of the shell and variance algebra is formalized in `SincProductShells` and `BirkhoffVariance`. The arbitrary-b Pochhammer grouping, integer-b ray law, and general covariance theorem above are new human-readable results and explicit Lean targets.

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17 Numerical evaluation of the finite-q spectrum

17.1 Chebyshev collocation

For smooth weights (in particular $q \geq 1$), the Perron root of (5.5) is efficiently obtained by Chebyshev–Lobatto collocation. Let

$$x_j = \frac{1 - \cos(j\pi/(N-1))}{2}, \qquad 0 \leq j < N,$$

interpolate f at the points $x_j/2$ and $(x_j+1)/2$ by the barycentric formula, and form the matrix representation of $\mathcal{L}_q$. Its leading eigenvalue converges rapidly to ρ_q.

For $q = 1$ the convergence is:

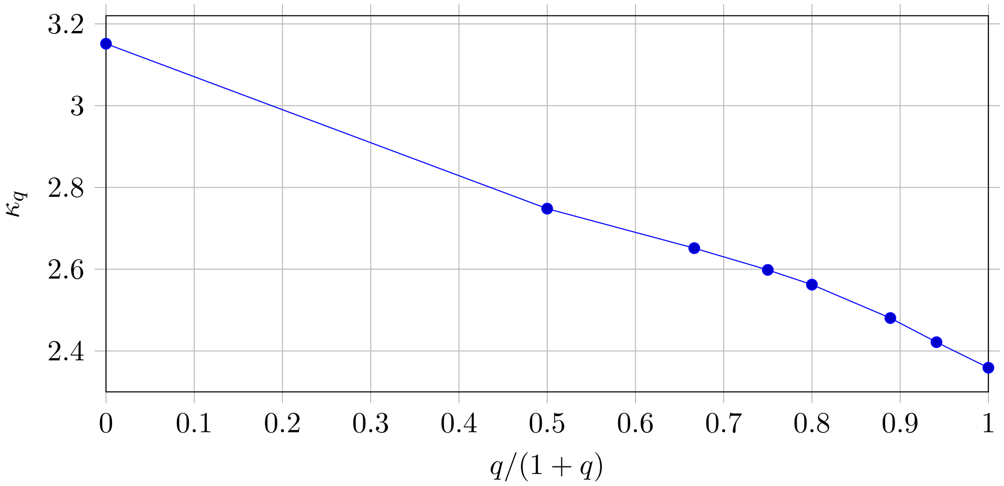
N	leading collocation eigenvalue
8	0.661322639955074
10	0.661322602233874
12	0.661322602061258
16	0.661322602060566
24	0.661322602060565

The rational enclosure in Theorem 9.1 is independent of this numerical convergence and provides a rigorous check on the computed value.

17.2 Sample spectrum

The following values illustrate the interpolation from typical behavior to the sharp uniform envelope. The rows $q = 0, 2, \infty$ are exact; the remaining rows are numerical collocation values.

q	ρ_q	$\Lambda_q = \rho_q^{1/q}$	κ_q
0	–	0.5000000000	3.1514961295
1	0.6613226021	0.6613226021	2.7480700149
2	0.5000000000	0.7071067812	2.6514961295
3	0.3948966324	0.7336593838	2.5983138062
4	0.3201941016	0.7522346457	2.5622414703
8	0.1612444589	0.7960412993	2.4805809434
16	0.0500719357	0.8293247927	2.4214870020
∞	–	0.8660254038	2.3590148791



The monotone curve is not a cosmetic refinement: choosing one row answers one specific notion of "amplitude." Moving from $q = 0$ to $q = \infty$ changes the power of x by almost 0.8, an effect much larger than any logarithmic correction.

18 Source-by-source reconciliation and correction of the draft

The pre-consolidation directory contained the original question, eight research reports, a gentle guide, two audits, and two script suites. Repetition was useful while the reports were independent; it is harmful in the final account because later corrections can leave an earlier theorem looking authoritative. This section records exactly what was retained, corrected, demoted, or generalized. The complete pre-consolidation evidence remains available in Git at commit `2e3567feb14947ee3ebcdab11adca64e746ad26f`.

18.1 Evidence-status legend

Six statuses are kept separate throughout this volume.

1. *Exact proof*: a complete argument is given here; many such results also have a named Lean declaration.
2. *Imported theorem*: the product-specific reduction is proved here, while a precisely stated classical theorem such as RPF, a martingale limit theorem, or Carleman–Hoischen approximation is cited.

3. *Rigorous certificate*: exact rational or interval arithmetic proves a stated enclosure, but does not validate every digit of a longer computed decimal.
4. *Numerical evidence*: stable computation or collocation supports a value or pattern, but supplies neither interval certification nor a proof of a limit.
5. *Frontier / conjectural*: a conjecture, proof architecture, or proposed strengthening is recorded with its missing bound or bridge stated explicitly.
6. *Lean-source-only*: current Lean source contains an exact declaration, but this volume has not yet supplied the corresponding conventional proof. Source presence and a fresh successful Lean build are reported separately.

In particular, a long stable decimal is not a rigorous enclosure, an exhibited eigenvalue is not automatically the second eigenvalue, and a formal Lean skeleton with an analytic hypothesis is not a proof of that hypothesis.

18.2 Source concordance

Source	Material retained	Adjudication in this volume
Original 2018 question [14]	Q2/Q3, finite Thue–Morse product-to-sum idea, reciprocal-Gamma and Pochhammer approach.	Questions restated precisely; formula (10) corrected in Theorem 3.1; false radius inference removed.
Report 1 [21]	Exact shell law, fixed rays, typical law, correct variance/CLT/LIL, integer-ratio generalization.	Dyadic core retained; integer-ratio result strengthened to Theorems 16.1 and 16.6.
Report 2 [22]	Calibrated Gelfond inequality, dense preimages, Lambert boundary layer, odd-multiple valleys.	Detailed extremal proofs retained in Sections 12 and 13 and theorem 14.1.
Report 3 [23]	First arithmetic-mean gauge and finite-q spectrum.	Mean theory retained conditional on RPF; inherited LIL constant π corrected to $\pi/\sqrt{2}$.
Report 4 [24]	Systematic transfer operator, exact $q = 0, 2$ rows, rational ρ_1 certificate, fixed-offset valleys.	Retained; its continuum of intermediate rates remains explicitly conditional.
Reports 5–6 [25, 26]	Broad reconciliations, clean Gordin proof, numerical spectrum, source audit.	Deduplicated into the main spine; their *adaptive gauge still open* status is superseded.
Report 7 [27]	Bounded-distortion obstruction and first shell-adaptive analytic gauge.	Strong obstruction restored; malformed CLT quantifier corrected; smoothing replaced by the stronger Report 8 lemma.
Report 8 [28]	Clean reciprocal construction, derivative-controlled flattening, analytic upgrade.	Canonical adaptive proof in Section 10.
Comparative audit [29]	LIL forensic audit, grid and energy identities, peak–area comparison, RMS eigen-identities.	Exact pieces proved in Theorems 4.3 and 8.3 to 8.5; spectral-rate and peak-CLT overclaims demoted.
Second-wave audit [30]	Corrected Q2 map, adversarial check of Reports 5–8, high-precision scripts.	Central logical corrections retained; its own appendix misprints are corrected below.
Gentle guide [31]	Accessible explanation of the log-square term, binary dynamics, four decay notions.	Intuition integrated into the introduction and proof commentary; unsupported numerical and *formula-like gauge* language removed.

18.3 Immutable source snapshot ledger

The twelve absorbed TeX sources remain available at pre-consolidation commit `2e3567feb14947ee3ebcdab11adca64e746ad26f`. To identify the precise inputs independently of their filenames, the following are SHA-256 digests of the TeX bytes extracted from that commit.

Source	SHA-256 of absorbed TeX
Original question	`62586a1f0e6d9a12a23bfd38c006365ffc0bf897e84c6b4d3d318ac6c0c29b62`
Report 1	`27a9fda6220d3e3cde110919c40f1946ccdeff1463e8ef7a5055fbb6c26ef7ed`
Report 2	`df24e59e68f1cacf0aa1c26f1f48b0e6bb2a643fccf8ac805a96b373a56576a8`
Report 3	`ae5ba2d44c565ce826471893cf867d6cf4c365a0a5f580fdd4548d0a2dedb73d`
Report 4	`b0a76b35fbb9a9af78c469f07d94eb4346bfaeace20cca193b3c9afa97ed6816`

Source	SHA-256 of absorbed TeX
Report 5	`12af56c41c2a2dad549bddf77477e17ec47cf7a5d2dc65df1ca96dd89880b455`
Report 6	`de912bde5096ff4e4db8b4ad3a176e3f7c35551688fcf4bb450437626619a924`
Report 7	`17426ff49f76d580c922045d693fb215552a51e68e5a45cff8442d5a46cf9ad9`
Report 8	`a1dcad9e686f4f0b39ef326c953501a3e03eef7fbd9b8c05351bbaaa6c211749`
Comparative audit	`871c3790ace2ce71bb906d17e96129aec79fcbb4fd069c412109b5da138e2ce7`
Second-wave audit	`e2aa812e12ba613cdb89a7bce56814aa034db99b4db68457e46848340f480684`
Gentle guide	`300cff33d98720436d655000b6d1070d1a86016e0ecc97235bc9e4b0a11f1791`

18.4 Why apparently different decay rates coexist

The power of x is determined by the numerator scale being sampled:

$$\frac{1}{2} \quad (q=0), \qquad \rho_1 \quad (q=1), \qquad \frac{1}{\sqrt{2}} \quad (q=2), \qquad \frac{\sqrt{3}}{2} \quad (q=\infty).$$

Substitution into $\kappa = \frac{1}{2} + \log_2(\pi/\Lambda)$ gives four different, simultaneously correct answers. The old draft's value $\log_2(2\pi)$ is exactly the $q = 2$ row.

The Lambert term is not a fifth power. It refines the $q = \infty$ scale when comparison is anchored at the left endpoint of a moving dyadic annulus. The exact $4/3$ ray is sharp relative to E_{κ_∞} at its own argument; the annular maximizer exploits the much larger deterministic baseline nearer the left endpoint. The fixed-offset and odd-multiple valleys are different again: they describe exceptional individual lobes and force zero liminf on the global-envelope scale.

18.5 Correction ledger

1. The published and Report 3 LIL constant π with denominator $\sqrt{n \log\log n}$ is too large by $\sqrt{2}$. The exact variance and Gordin decomposition prove $\pi/\sqrt{2}$; see Theorems 7.1 and 7.2.

2. The original Pochhammer formula needs a negative exponent, denominator $1-4^{-k}$ independent of m, a sum beginning at $k = 1$, and consistent prefix indices. A convergence radius never implies that the represented entire function vanishes outside it.

3. The natural-gauge obstruction is not an impossibility theorem for all analytic or all elementary functions. It excludes stabilizing and bounded-distortion shell shapes. The analytic adaptive gauge in Theorem 10.3 proves unrestricted Q2 affirmatively.

4. The implication *Hardy-field, hence elementary* is invalid. Elementary oscillatory compositions can leave every Hardy field; the bounded oscillatory example before Section 10 is handled instead by bounded distortion.

5. The Gelfond estimate is sharp in exponential rate. The lower orbit attains $(\sqrt{3}/2)^n$; the convenient upper bound with one fewer power is not itself attained for every n.

6. The eigen-identities at $-1/4$ are exact, but the *second eigenvalue* claim and the $(-1/2)^n$ relative RMS rate were inferred from collocation. They are not theorem statements here.

7. The limiting arithmetic profile is continuous and driven by the distribution function of an atomless singular measure. Division by the mantissa contributes an ordinary absolutely continuous factor, so the profile itself is not a singular distribution function or a measure.

8. Complete logarithmic shells have masses converging to one common limit; they are not exactly equal at finite level.

9. Finite-grid profile calculations suggest excursions at least as far as roughly 0.91627 and 1.11814, but no interval-arithmetic argument certifies the global extrema of the limiting profile.
10. The comparative audit's peak CLT and limiting peak/area ratio require a certified cell-to-maximum transfer. The rigorous peak–area inequality is retained, but the limit $1.92372\ldots$ remains numerical evidence.

18.6 Correct constants and their status

The exact closed forms give

$$\begin{aligned}\kappa_0 &= 3.151496129472318798043279\ldots,\\ \kappa_2 &= 2.651496129472318798043279\ldots,\\ \kappa_\infty &= 2.359014879111740707316409\ldots.\end{aligned}$$

High-precision collocation, stable to substantially more digits than displayed but not an interval proof, gives

$$\begin{aligned}\rho_1 &= 0.661322602060565615073529 12\ldots,\\ \kappa_1 &= 2.748070014871332673917143 56\ldots,\\ A_1 &= 0.091266124131588212286563 07\ldots,\\ B_1 &= 0.065059235040376921430990 53\ldots,\\ A_1^{\log} &= 0.093860635756781263540683 41\ldots.\end{aligned}$$

The rigorous enclosures remain the much wider published and exact-rational intervals in Theorem 9.1. In particular, the longer decimals must not be cited as certified.

The second-wave appendix itself had four unnoticed transcription errors: the tails of $\kappa_0, \kappa_2, \kappa_\infty$ were printed incorrectly, and ρ_8 was printed as 0.161244458899776. Its own bundled spectral data instead stabilize at

$$\rho_8 = 0.161244458899778546776960 7\ldots.$$

18.7 The literal Cesaro question, finally

The natural gauge

$$g_1(x) = A_1 x^{-\kappa_1} \exp\left(-\frac{(\log x)^2}{2\log 2}\right)$$

is positive, analytic, eventually decreasing, and has alternating derivative signs eventually for each fixed derivative order. It solves Q3, and Q2 along dyadic endpoints. For unrestricted endpoints it has the nonconstant profile (9.12). No bounded-distortion shell correction can flatten that profile. The shell-adaptive gauge g_{ad} escapes precisely by developing unbounded, finer-and-finer microstructure and solves the literal unrestricted Q2. If one insists on a fixed nonadaptive scale, the logarithmic mean in Theorem 11.1 is the canonical all-endpoint repair.

18.8 Imported inputs and formalization boundary

The product algebra, zero divisor, Thue–Morse sign, exact factorization, finite variance, finite-grid Parseval, Gelfond inequality, Lambert optimization, peak–area estimate, and both valley theorems are proved here. The exact L^2 row is finite Fourier algebra. Three classical analytic packages remain imported with their hypotheses visible:

1. Ruelle–Perron–Frobenius quasi-compactness and simplicity for (5.5);
2. stationary reverse-martingale CLT/LIL theorems;

3. Carleman–Hoischen simultaneous approximation of a C^1 function and its derivative.

The Collatz–Wielandt/Sturm appendix gives an exact machine-checkable enclosure for ρ_1. The verification scripts supply reproducible numerical evidence for the longer constants and finite-level profiles, but they do not change that evidence category. The Lean cross-reference in Section C records, theorem by theorem, which exact substrate is already represented by exact Lean declarations and which analytic bridge remains a frontier.

19 Conclusion

The Fourier image of Rvachev's up-function has one deterministic core and several genuinely different stochastic or extremal refinements. The exact identity

$$|\Phi(2^n y)| = |\Phi(y)|\pi^{-n}y^{-n}2^{-n(n+1)/2}\prod_{j=1}^{n}|\sin(\pi 2^j y)|$$

separates them perfectly. The triangular denominator produces $-(\log x)^2/(2\log 2)$. The lacunary numerator then selects a pressure level: Lebesgue geometric mean, Perron L^1 mean, Parseval L^2 mean, or maximizing orbit.

The final quantitative dictionary is:

$$\begin{aligned}
\kappa_0 &= 3.1514961294723187980\ldots && \text{typical / geometric mean,}\\
\kappa_1 &= 2.7480700148713326739\ldots && \text{arithmetic mean and Cesaro gauge,}\\
\kappa_2 &= 2.6514961294723187980\ldots && \text{RMS; the old draft's exponent,}\\
\kappa_\infty &= 2.3590148791117407073\ldots && \text{sharp pointwise upper envelope.}
\end{aligned}$$

The other three rows have exact closed forms; the displayed κ_1 digits are spectral computation accompanied by a separate rigorous enclosure. Typical logarithms fluctuate as $(\pi/\sqrt{2}+o(1))\sqrt{(\log x/\log 2)\log\log\log x}$ along the LIL extremal subsequences. Raw dyadic-annulus maxima carry an additional Lambert-W term of order $-(\log\log x)^2/(2\log 2)$. Fixed-offset lobe maxima after powers of two can instead have the doubled leading coefficient $-(\log x)^2/\log 2$.

For the question that originally motivated the investigation, the positive analytic gauge $A_1E_{\kappa_1}$ gives the requested two-sided bounded Cesaro normalization and the exact dyadic-endpoint limit. Its failure at unrestricted endpoints, and the stronger bounded-distortion obstruction, are structural: the last dyadic shell retains a singular distribution of mass. They do not contradict existence. The adaptive gauge g_{ad} develops the necessary finer-and-finer shell microstructure and solves the literal unrestricted analytic problem. Logarithmic averaging gives a simpler, fixed-scale all-endpoint alternative.

The corrected Pochhammer theorem recovers the useful part of the original attempted answer without confusing a local convergence disk with an asymptotic law. Its arbitrary-radius and integer-ratio extensions show that the dyadic case belongs to a broader family with

$$\kappa_{\mathrm{typ}}(b) = \frac{1}{2}+\log_b(2\pi), \qquad \sigma_b^2 = \frac{\pi^2}{12}\frac{b+1}{b-1}.$$

Thus what looked like one irregular asymptotic is a compact pressure spectrum, one Lambert boundary layer, several arithmetic valley mechanisms, and a clean geometric-ratio universality class. Every surviving claim now carries its proof status and source provenance explicitly.

A Exact Collatz–Wielandt certificate for ρ_1

The following SymPy script uses only exact rational arithmetic. It constructs the rational function in the proof of Theorem 9.1; Sturm's theorem, invoked by `count_roots`, certifies that the relevant numerators and denominators have no roots on $[0, 1]$. Positivity at $t = 0$ therefore proves positivity throughout the interval.

```
import sympy as sp

t = sp.symbols("t", real=True)
u = 2*t/(1+t**2)
v = (1-t**2)/(1+t**2)

c = sp.Rational(376189, 10**6)
d = sp.Rational(-69093, 10**6)
e = sp.Rational(15483, 10**6)
F = lambda z: 1 + c*z + d*z**2 + e*z**3

R = sp.cancel((u*F(u) + v*F(v)) / (2*F(2*u*v)))
lo = sp.Rational(66126807, 10**8)
hi = sp.Rational(66134891, 10**8)

for expr in (R - lo, hi - R):
    num, den = map(lambda p: sp.Poly(p, t),
                   sp.together(expr).as_numer_denom())
    assert num.eval(0) > 0 and den.eval(0) > 0
    assert num.count_roots(0, 1) == 0
    assert den.count_roots(0, 1) == 0

print("Certified:", lo, "< rho_1 <", hi)
```

For the high-precision decimal, one may replace the exact certificate by a Chebyshev collocation matrix. The rapid stabilization displayed in Section 17 is typical for this analytic weighted-composition operator.

B Notation cross-reference

Symbol	Meaning
$\Phi = \widehat{\mathrm{up}}$	Fourier transform of Rvachev's up-function
a	$\log 2$
T	doubling map $x \mapsto 2x \pmod 1$
$s(x)$	$\lvert\sin(\pi x)\rvert$
$B_n(y)$	$\prod_{j=1}^{n} s(T^j y)$
$E_\kappa(x)$	$x^{-\kappa} \exp(-(\log x)^2/(2\log 2))$
$W_\kappa(y)$	bounded dyadic phase factor in (3.12)
$\mathcal{L}_q$	weighted Perron operator (5.5)
ρ_q	Perron root of $\mathcal{L}_q$
Λ_q	$\rho_q^{1/q}$, the L^q numerator growth per dyadic level
κ_q	sharp power in the normalized L^q shell decay
M_m	unique maximum of $\lvert\Phi\rvert$ on $[m, m+1]$

C Lean declaration crosswalk and parity ledger

This appendix enforces the intended two-way documentation discipline. It lists each substantive theorem family used in this volume, not every implementation lemma. An *exact* row means that current Lean source contains a theorem with the displayed mathematical content; an *abstract* row means Lean proves the deduction from explicit hypotheses but not the analytic hypothesis itself.

The present document build does not by itself constitute a fresh aggregate Lean compilation, so compilation evidence must be reported separately from this source crosswalk.

C.1 Human-readable results with exact Lean counterparts

The middle column names declarations, not merely files. All are in namespace `Fabius`. Mixed results are split so a Lean declaration is never made to cover a strictly stronger human theorem.

Human-readable result	Exact Lean declaration(s)	Location in this volume
Entire sinc product, evenness, real-axis reality, value at zero, and the classical Fourier-product bridge	`rvachevFourierProduct_differentiable`, `rvachevFourier_neg`, `rvachevFourier_ofReal_im_eq_zero`, `rvachevFourier_zero`, `rvachevFourier_eq_product`	Sections 1 and 3
Exact nonzero complex zero set and analytic multiplicity $1 + \nu_2(m)$	`rvachevFourierProduct_eq_zero_iff`, `analyticOrderAt_rvachevFourierProduct_int`	Theorem 3.3
Exact zero-safe dyadic shell factorization and exponent algebra	`norm_rvachevFourierProduct_two_pow_mul_cross`, `norm_rvachevFourierProduct_two_pow_mul`, `shell_exponent_identity`	Theorem 3.4
Finite Thue–Morse product-to-sum identity	`prod_sin_two_pow_eq_thueMorse_sum`	Equations (3.13) and (3.14)
Strict central/side-lobe concavity, unique lobe maxima, and Thue–Morse lobe sign	`strictConcaveOn_central_log_series`, `strictConcaveOn_lobe_log_series`, `existsUnique_isMaxOn_rvachevFourier_lobe`, `rvachevFourier_eq_thueMorse_sign_mul_norm`	Theorem 4.1
Sharp Gelfond inequality and equality orbit	`abs_prod_sin_two_pow_le_sharp`, `abs_prod_sin_two_pow_third`	Theorem 12.2
Global envelope, exact fixed ray, and failure of every stronger exponent	`norm_rvachevFourier_le_decayGauge`, `norm_rvachevFourier_peak_ray_envelope`, `not_exists_norm_rvachevFourierProduct_le_decayGauge_of_kappaInf_lt`	Theorems 12.3 and 12.4
Exact κ dictionary, ordering, and closed-form interval certificates	`kappaOf_half`, `kappaOf_sqrt_two`, `kappaOf_sqrt_three`, `kappaInf_lt_kappa_two`, `kappa_two_lt_kappa_zero`, `kappa_zero_bracket`, `kappa_two_bracket`, `kappa_inf_bracket`	Sections 1.1 and 12
Continuous quadratic lacunary integral and dyadic-grid Parseval	`integral_prod_sin_sq_two_pow`, `sum_prod_sin_sq_two_pow`	Equation (8.3) and theorem 8.3
Root-of-unity sine product	`prod_two_sin_pi_div_two_pow`	Theorem 8.3
Exact cocycle covariance, finite Birkhoff variance, and product bridge	`integral_cocycle_covariance`, `integral_sq_birkhoff_sum_all`, `integral_sq_log_prod_two_sin_all`	Theorem 7.1

Human-readable result	Exact Lean declaration(s)	Location in this volume
Gordin pointwise algebra and the $\pi^2/12, \pi^2/4$ Parseval values	`two_log_two_sin_sub`, `log_abs_tan_half_add`, `integral_sq_log_two_sin_pi_mul`, `integral_sq_log_abs_tan_pi_mul`	Section 7
Explicit $q = 2$ transfer eigen-identities	`rms_transfer_const_eigen`, `rms_transfer_sin_eigen`, `rms_transfer_one_add_cos_eigen`	Theorem 8.4
Plancherel and sinc-square energy identities (not the half-integer sampling formula)	`integral_norm_sq_rvachevFourier_eq_two_mul_fabiusSquareEnergy`, `fabiusSquareEnergy_eq_integral_Ioi_tprod_sinc_sq`	Theorem 8.5
Global dyadic Pochhammer and reciprocal-Gamma products (not the extracted tail series)	`rvachevFourierProduct_eq_tprod_complexQPochhammerInf`, `dyadicReciprocalGamma_mul_neg`, `rvachevFourierProduct_eq_one_div_dyadicGamma_mul`	Theorems 3.1 and 16.1
Existence and uniqueness of the transfer-supremum growth root (not yet its RPF spectral identification)	`exists_perron_exponent`, `exists_perron_root`, `existsUnique_perron_root`	Theorems 5.1 and 9.1
Lean Bernstein enclosures of the growth root and its κ image	`perron_bracket_pointwise`, `perron_root_enclosure`, `kappa_one_enclosure`	Separate-certificate remark after Theorem 9.1

C.2 Exact Lean-source-only results still owed a conventional proof here

These declarations are exact source statements, but the present volume uses them only as context or supporting infrastructure and does not yet reproduce their full conventional proofs. They are therefore explicit documentation debts, not entries in the preceding human-proof parity table.

Mathematical content	Exact Lean declaration(s)
Rapid real-axis decay of the Fourier transform and all of its iterated derivatives	`rvachevFourier_real_rapidDecay`, `rvachevFourier_real_iteratedDeriv_rapidDecay`
Typical logarithmic exponent in measure and uniform $\sqrt{n}$-scale tightness	`measureReal_log_prod_ge_le`, `tendsto_measureReal_log_prod_ge`, `measureReal_log_prod_sqrt_ge_le`, `integral_log_prod_two_sin`, `integral_abs_log_prod_le`
Baseline $1/(\pi\|x\|)$ bound, global one-bound, and strictness away from zero	`norm_rvachevFourier_le_inv`, `norm_rvachevFourier_le_one`, `norm_rvachevFourier_lt_one_of_ne_zero`, `norm_rvachevFourier_eq_one_iff`

C.3 Abstract Lean deductions whose analytic premises remain visible

Deduction formalized	Exact Lean declaration(s)	Missing product-specific input
Eventual two-sided Cesaro boundedness from normalized shell bounds	`eventually_bounded_of_monotone_of_dyadic`	The $q = 1$ RPF shell asymptotic
Dyadic profile aggregation from complete- and partial-shell limits	`tendsto_cumulative_div_two_pow`, `tendsto_cumulative_profile`	Eigenmeasure convergence, atomlessness, and singularity
Logarithmic mean from convergent shell masses and bounded endpoint errors	`tendsto_sum_add_bdd_div`	The concrete weighted shell limit
Partial-shell linearity forced by an unrestricted additive limit	`tendsto_shell_increment`	The singular-measure contradiction for the sinc product
Finite-partition measure equalization and the diagonal selection mechanism	`setIntegral_equalizer`, `abs_setIntegral_equalizer_sub_le`, `exists_diagonal_tendsto_zero`	Smooth bridges, global gluing, and Carleman–Hoischen approximation

C.4 Human theorems still awaiting full Lean formalization

The remaining frontier is now explicit rather than hidden in prose:

1. the full RPF eigenfunction/eigenmeasure/spectral-gap package for (5.5), including the singular full-support $q = 1$ eigenmeasure;
2. the stationary martingale CLT and LIL applications, although their exact variance and coboundary algebra are formal;
3. the bounded-distortion obstruction and the completed smooth/analytic adaptive gauge;
4. the Lambert annular boundary layer and both complete peak-valley asymptotics, beyond the already formal local zero orders and global envelope;
5. the quantitative peak–area estimate and any eventual peak-height CLT;
6. the first-zero-extracted and arbitrary-J Pochhammer series;
7. the integer-b shell law, typical exponent, covariance, variance, and general martingale decomposition from Section 16;
8. the half-integer sampling equality in Theorem 8.5, if not yet exported as a named theorem by the energy module.

Every theorem promoted in the prose now appears in the exact, abstract, or unformalized ledger above, while the three exact Lean-source-only families are visibly recorded as documentation debts. Any future substantive Lean theorem added to the facade should add a row here; any future theorem promoted in the prose should name its declaration or remain visibly marked as imported, numerical, or frontier work.

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